

TOWER CRANE CLIMBING

BC REGULATIONS & RED SEAL

WORKSAFEBC ENFORCED PROCEDURES

FOR TOWER CRANE CLIMBING OPERATIONS



**WORKSAFEBC
COMPLIANT**



**SAFE WORK
EVERYTIME**



**PLAN • INSPECT
EXECUTE • VERIFY**



**RED SEAL
EXAM FOCUSED**



**COMPLY WITH WORKSAFEBC
PART 14 – CRANES & HOISTS**



**QUALIFIED SUPERVISOR
DIRECTS THE CLIMB**



**ENGINEERED SUPPORT
FOUNDATIONS, TIE-INS, SHORING**



**FOLLOW MANUFACTURER
PROCEDURES FIRST**

CLIMBING TERMINOLOGY

KEY COMPONENTS & DEFINITIONS

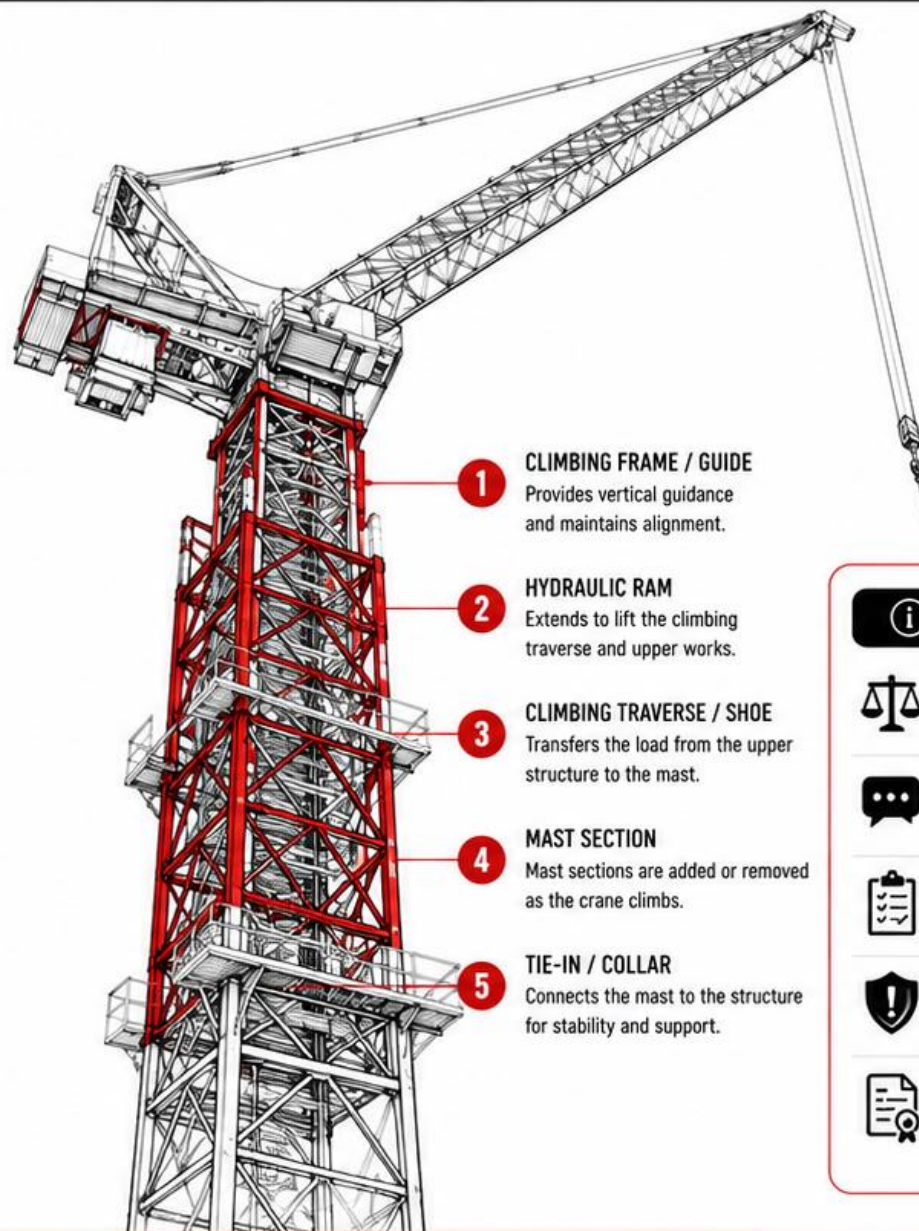
Understanding climbing terminology is essential for safe communication, preparation, and execution of tower crane climbing operations.

- | | | |
|---|---------------------------------|--|
| 1 | CLIMBING FRAME / GUIDE | Steel frame that surrounds the mast and provides guidance and support during climbing. |
| 2 | HYDRAULIC RAM | Hydraulic cylinder that pushes the climbing traverse (shoe) up the mast. |
| 3 | CLIMBING TRAVERSE / SHOE | The horizontal beam/shoe that bears against the mast and is lifted by the hydraulic ram. |
| 4 | MAST SECTION | Standard tower sections that are added or removed during the climbing sequence. |
| 5 | TIE-IN / COLLAR | Structural connection between the crane mast and the building or structure. |



KEY TAKEAWAY

Every component has a specific role during the climbing sequence. Know the names. Know the purpose. Follow the plan.



ALSO IMPORTANT



BALANCE

Proper balance is critical throughout the climb.



COMMUNICATION

Clear, single-point communication at all times.



SEQUENCE

Follow the manufacturer's approved climbing sequence.



HAZARDS

Recognize and control hazards before, during, and after climbing.



CERTIFICATION

Climb must be verified, certified, and documented before return to service.

THE CLIMBING PROCESS



PLAN

Plan the climb.



PREPARE

Verify supports, power, and crew.



INSPECT

Inspect equipment and components.



BRIEF

Review the plan with the team.



BALANCE

Balance the crane as required.



CLIMB

Execute the climbing sequence.



SECURE

Secure all connections and components.



CERTIFY

Obtain certifications, adjust limits, test.



RETURN TO SERVICE
Crane is tested and returned to service.

TOP CLIMBING vs. BOTTOM CLIMBING

TWO METHODS – ONE GOAL: SAFE HEIGHT INCREASE

Tower cranes can be climbed using two methods depending on the crane design, site conditions, and manufacturer specifications.

TOP CLIMBING



- ✓ Climbing frame is attached to the top of the mast.
- ✓ Upper works remain at the top while the mast grows from below.
- ✓ Common on luffing jib tower cranes.
- ✓ Requires balance during the climb.

BOTTOM CLIMBING



- ✓ Mast sections are added above the turntable.
- ✓ Upper works climb with the mast.
- ✓ Common on hammerhead tower cranes.
- ✓ Requires balance during the climb.



IMPORTANT | Always follow the manufacturer's climbing procedure. Never change the sequence or method.



A QUALIFIED SUPERVISOR MUST DIRECT ALL CLIMBING OPERATIONS.

TOP CLIMBING – TYPICAL SEQUENCE

1

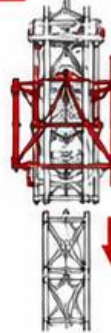

Climbing frame engaged. Crane balanced and ready.

2


Hydraulic ram extends, lifting the upper works via the climbing traverse/shoes.

3


Clearance created for new mast section.

4


New mast section installed and secured.

5


Hydraulic ram retracts, setting the upper works down onto the new section.

6


Connections secured. Climb complete. Proceed to next tie-in as per plan.



REMEMBER

- Balance the crane as specified by the manufacturer.
- Never remove or install any pin or connection out of sequence.
- Maintain constant communication with the climbing supervisor.
- Stop immediately if any abnormal condition occurs.

FACTORS THAT AFFECT CLIMBING



Wind



Temperature



Freezing Rain / Ice



Snow



Lightning



Fog / Visibility

THE CLIMBING PRINCIPLES



PLAN

Plan every climb.



PREPARE

Verify supports, power, and crew.



INSPECT

Inspect equipment and components.



BRIEF

Review the plan with the team.



BALANCE

Balance the crane as required.



CLIMB

Execute in sequence. No deviations.



SECURE

Secure all connections and components.



CERTIFY

Obtain certifications and adjust limits.



RETURN TO SERVICE

Test and return to service.

WORKSAFEBC LEGAL REQUIREMENTS

WHAT THE LAW REQUIRES BEFORE, DURING, AND AFTER CLIMBING

Employers, supervisors, and workers must comply with the Occupational Health and Safety Regulation (OHSR) Part 14.

BEFORE THE CLIMB



- **Notice of Project – Tower Crane (NOP-TC)**
Submit to WorkSafeBC at least 2 weeks before the climb (OHSR 14.72).
- **Qualified Persons**
Climbing must be performed by qualified persons following the manufacturer's instructions or engineer's instructions when a variation is used (OHSR 14.73.2).
- **Qualified Supervisor**
A qualified supervisor must be in place and direct the climbing activity (OHSR 14.73.2).
- **Engineering Support**
Foundations, ties, shoring and bracing must be professionally engineered (OHSR 14.74).

DURING THE CLIMB



- **Follow the Plan**
Follow the approved climbing plan and manufacturer's specified sequence.
- **Maintain Balance**
Keep the crane balanced as specified by the manufacturer.
- **Communication**
Maintain clear, single-point communication with the climbing supervisor.
- **Stop If Unsafe**
Stop immediately if any abnormal condition, loss of balance, or deviation from the plan occurs.

AFTER THE CLIMB



- **Verification & Certification (OHSR 14.75)**
Qualified supervisor must verify the climb was completed per manufacturer's or engineer's specifications.
- **Engineer Certification (if applicable)**
Engineer must certify any affected connections, bracing or shoring.
- **Overload System Adjustment**
A qualified person must adjust the overload prevention system as necessary.
- **Climbing Report**
Make the required report available at the workplace, including limit settings and counterweight/ballast weights.

NOP-TC REQUIRED INFORMATION

AT LEAST 2 WEEKS BEFORE THE CLIMB

- ✓ Site location and project name
- ✓ Crane make, model, and serial number
- ✓ Type of activity: Climbing (increase in height)
- ✓ Start date of climb
- ✓ Name and contact of employer responsible
- ✓ Name of qualified supervisor
- ✓ Confirmation of site-specific safe work procedure
- ✓ Tie-in/shoring engineered status
- ✓ Any significant changes must be reported



REMEMBER

Compliance with the law is the foundation.
Manufacturer first. Engineer where a variation is approved. Safety is never optional.

THE CLIMBING PROCESS



PLAN

Review plan, drawings, and engineered data.



PREPARE

Verify supports, power, grounding, and equipment.



INSPECT

Inspect climbing equipment and connections.



BRIEF

Review sequence, roles, hazards, and comms.



BALANCE

Balance crane as specified by the manufacturer.



CLIMB

Execute sequence step-by-step.



SECURE

Secure all pins, connections, and tie-ins.



CERTIFY

Obtain certifications, adjust limits, and document.



RETURN TO SERVICE

Test crane and return to normal operations.



QUALIFIED SUPERVISOR – ROLES & RESPONSIBILITIES

DIRECTS THE CLIMB. ENSURES COMPLIANCE. PROTECTS EVERYONE.

WorkSafeBC requires a Qualified Supervisor to be in place for all tower crane climbing activities. The supervisor must be present, competent, and have the authority to stop the work if safety is at risk.



WHO IS A QUALIFIED SUPERVISOR?

- ✓ Has knowledge, training, and experience specific to the crane make/model and climbing activity.
- ✓ Understands manufacturer's climbing procedures and applicable regulations.
- ✓ Has successfully supervised similar climbing operations.
- ✓ Is recognized by the employer and has the authority to direct and stop work.



THE SUPERVISOR MUST:

- Be on site and actively supervise the climb.
- Review and confirm the climbing plan with the crew.
- Verify workers are qualified for their tasks.
- Ensure communication is clear and effective.
- Ensure the plan and sequence are followed.
- Stop the climb if anything is unsafe.



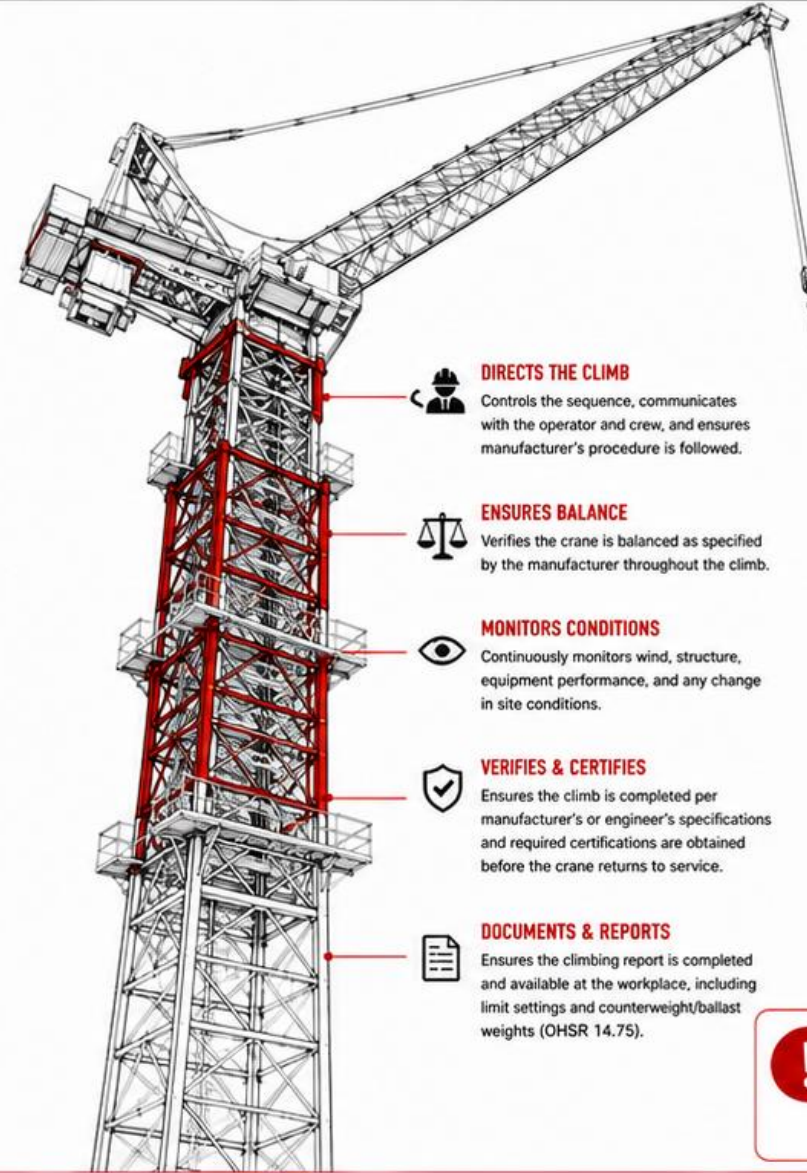
SUPERVISOR CHECKLIST

- ☐ Climbing plan reviewed and understood
- ☐ Manufacturer's instructions on site
- ☐ NOP-TC submitted (**≥ 2 weeks before climb**)
- ☐ Site-specific safe work procedure in place
- ☐ Erection crew qualified and briefed
- ☐ Climbing frame inspected and approved
- ☐ Mast sections and pins verified
- ☐ Hydraulic system checked (pressure, hoses, leaks)
- ☐ Power supply and grounding verified
- ☐ Tie-ins / shoring / building support ready
- ☐ Exclusion zone established and controlled
- ☐ Weather conditions within manufacturer limits
- ☐ Emergency / rescue plan in place
- ☐ Communication system tested and confirmed
- ☐ All crew briefed – roles and sequence
- ☐ Last-minute walk-through completed



SUPERVISOR AUTHORITY

If you see something unsafe – **STOP THE CLIMB.**
 No climb is more important than safety.



DIRECTS THE CLIMB

Controls the sequence, communicates with the operator and crew, and ensures manufacturer's procedure is followed.



ENSURES BALANCE

Verifies the crane is balanced as specified by the manufacturer throughout the climb.



MONITORS CONDITIONS

Continuously monitors wind, structure, equipment performance, and any change in site conditions.



VERIFIES & CERTIFIES

Ensures the climb is completed per manufacturer's or engineer's specifications and required certifications are obtained before the crane returns to service.



DOCUMENTS & REPORTS

Ensures the climbing report is completed and available at the workplace, including limit settings and counterweight/ballast weights (OHSR 14.75).



SUPERVISOR SKILLS

- ✓ Technical knowledge
- ✓ Climbing experience
- ✓ Leadership
- ✓ Communication
- ✓ Problem solving
- ✓ Decision making
- ✓ Risk management
- ✓ Attention to detail



REMEMBER

A ticket does not automatically make someone a Qualified Supervisor. Competence must be demonstrated for the specific crane and activity.

THE CLIMBING PROCESS



PLAN

Review plan, drawings, and engineered data.



PREPARE

Verify supports, power, grounding, and equipment.



INSPECT

Inspect climbing equipment and connections.



BRIEF

Review sequence, roles, hazards, and comms.



BALANCE

Balance crane as specified by the manufacturer.



CLIMB

Execute sequence step-by-step.



SECURE

Secure all pins, connections, and tie-ins.



CERTIFY

Obtain certifications, adjust limits, and document.



RETURN TO SERVICE

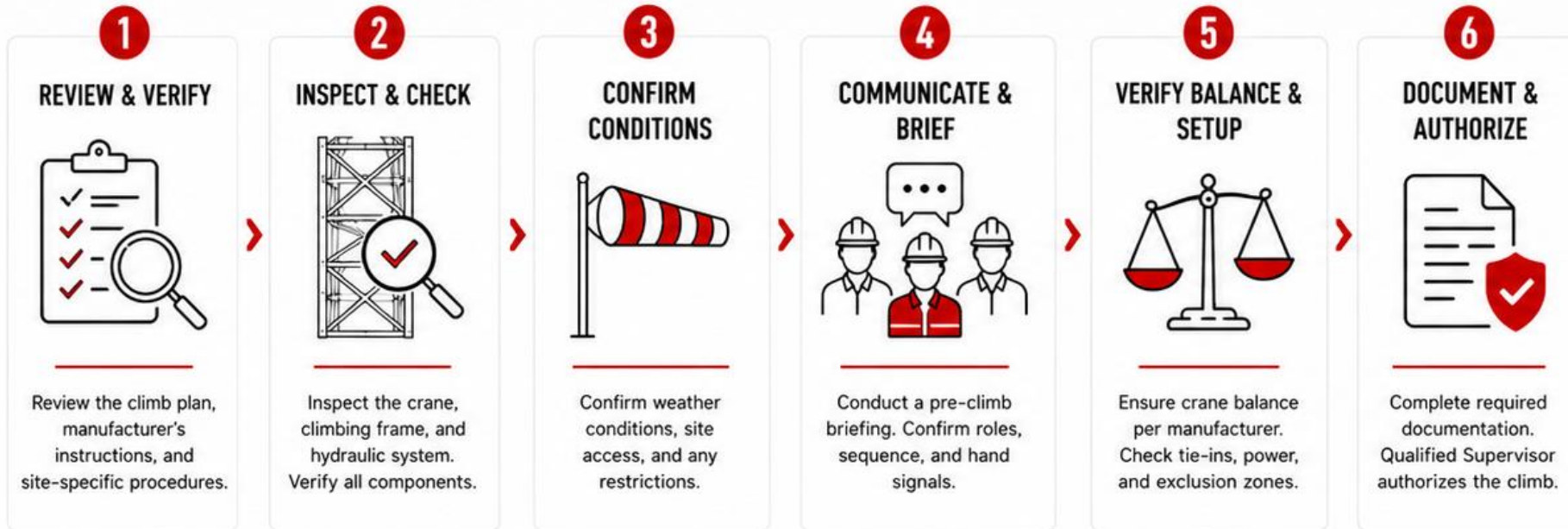
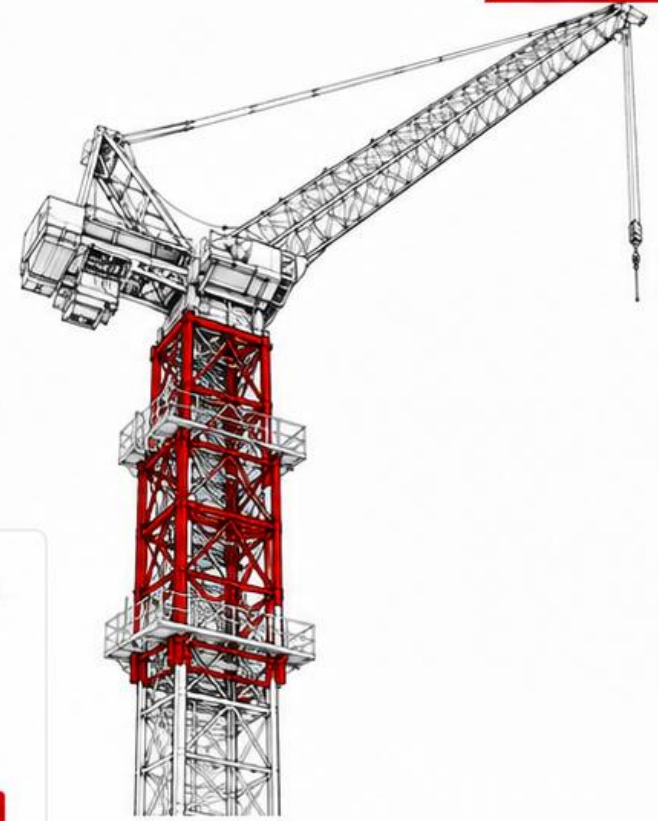
Test crane and return to normal operations.



PRE-CLIMB PLANNING

PLAN IT. VERIFY IT. CLIMB IT.

A successful climb starts long before anyone leaves the ground.
Follow a structured pre-climb planning process.



KEY PRINCIPLE

Good planning eliminates surprises and creates the conditions for a safe, controlled climb.

THE CLIMBING PROCESS



PLAN
Pre-climb planning



PREPARE
Set up & prepare



INSPECT
Inspect & verify



BRIEF
Team briefing & assign roles



BALANCE
Check balance & conditions



CLIMB
Execute climb sequence



SECURE
Secure connections & components



CERTIFY
Complete documentation



RETURN TO SERVICE
Test & return to normal operations

PREPARE THE CRANE & SITE

GET EVERYTHING READY. MINIMIZE RISK.

Preparation ensures the climb can be completed safely, efficiently, and in the correct sequence.
 Nothing gets lifted until everything is ready.


1

STAGE CLIMBING EQUIPMENT



- Climbing frame, brackets, and pins
- Tools and hardware
- Tag lines and lifting gear

2

VERIFY MAST COMPONENTS



- Correct mast sections and orientation
- Pins, bolts, and hardware
- Clean and ready to install

3

PREPARE POWER & HYDRAULIC SYSTEMS



- Power supply isolated and locked out
- Hydraulic lines inspected
- Adequate fluid level

4

TIE-INS & SHORING READY



- Tie-in points identified
- Shoring/temporary supports prepared
- Hardware inspected

5

ESTABLISH EXCLUSION ZONE



- Barricades and signage in place
- Area below crane secured
- Access controlled

6

ACCESS & FALL PROTECTION



- Safe access to climbing position
- Fall protection in place
- Anchor points verified

7

RESCUE PLAN & COMMUNICATION



- Rescue plan in place
- First aid available
- Radios tested and working

THE CLIMBING PROCESS


PLAN
 Pre-climb planning

PREPARE
 Set up & prepare

INSPECT
 Inspect & verify

BRIEF
 Team briefing & assign roles

BALANCE
 Check balance & conditions

CLIMB
 Execute climb sequence

SECURE
 Secure connections & components

CERTIFY
 Complete documentation

RETURN TO SERVICE
 Test & return to normal operations

INSPECT EVERYTHING

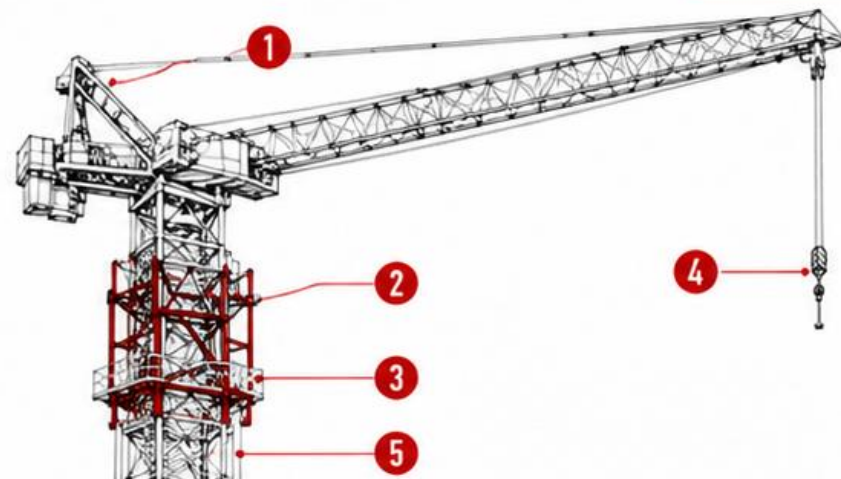
FIND IT. FIX IT. VERIFY IT.

A thorough inspection identifies issues before they become hazards. Inspect all components, systems, and equipment per manufacturer instructions and site procedures.



IF IN DOUBT – TAG IT OUT

Do not use any component, tool, or system that is damaged, defective, or not verified. Report and resolve before proceeding.


1

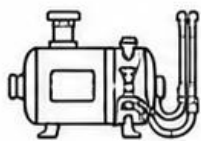
CRANE & STRUCTURE



- Jib, counter-jib, and connections
- Slew ring and bearing
- Hoist, trolley, and hooks
- Wire ropes and reeving
- Limit switches and safeties
- Structural members and bolts
- Climbing frame and brackets

2

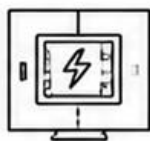
HYDRAULIC SYSTEM



- Hoses, fittings, and cylinders
- Leaks, damage, and wear
- Fluid level and condition
- Pressure gauges
- Valves and controls
- Function test all operations

3

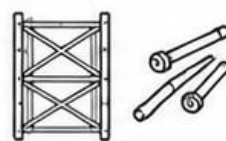
ELECTRICAL SYSTEM



- Power supply and cables
- Connections and breakers
- Control system and wiring
- Emergency stop
- Grounding and bonding
- Test all safety circuits

4

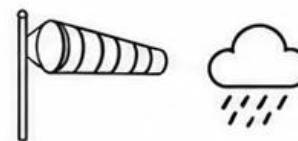
CLIMBING EQUIPMENT



- Climbing frame and rams
- Pins, bolts, and locking devices
- Brackets and tie-in hardware
- Wear, cracks, deformation
- Tools and lifting devices
- Tag lines and lifting gear

5

SITE CONDITIONS



- Wind within manufacturer limits
- Weather and visibility
- Tie-in/shoring points ready
- Surface and bearing capacity
- Housekeeping and access
- Exclusion zone in place

6

DOCUMENT & VERIFY



- Record inspection results
- Note deficiencies and actions
- Verify repairs completed
- Supervisor signs off
- Retain records

THE CLIMBING PROCESS


PLAN
 Pre-climb planning

PREPARE
 Set up & prepare

INSPECT
 Inspect & verify

BRIEF
 Team briefing & assign roles

BALANCE
 Check balance & conditions

CLIMB
 Execute climb sequence

SECURE
 Secure connections & components

CERTIFY
 Complete documentation

RETURN TO SERVICE
 Test & return to normal operations


CREW BRIEFING

SHARED PLAN. CLEAR ROLES. SAFE CLIMB.

A thorough briefing ensures everyone understands the plan, their role, and the hazards. No climb starts without a brief.



TEAM COMMUNICATION

- Ask questions.
- Confirm understanding.
- Speak up about concerns.
- Stop the work if anything is not safe.



KEY TOPICS TO COVER

1

CLIMB OVERVIEW



- What we are doing
- Sequence and method
- Start and finish points

2

ROLES & RESPONSIBILITIES



- Climbing Supervisor
- Crane Operator
- Riggers / Installers
- Signal Person

3

COMMUNICATION PLAN



- Hand signals / radio
- Call signs
- Who communicates with the operator

4

HAZARDS & CONTROLS



- Fall hazards
- Dropped objects
- Pinch points
- Electrical / hydraulic hazards

5

EXCLUSION ZONE



- Zone established
- Access controlled
- Warning signage in place

6

EMERGENCY PLAN



- Rescue plan and location
- First aid available
- Emergency contacts
- Stop work authority

7

CONFIRM UNDERSTANDING



- Questions?
- Concerns?
- Everyone confirms they understand

THE CLIMBING PROCESS



PLAN
Pre-climb planning



PREPARE
Set up & prepare



INSPECT
Inspect & verify



BRIEF
Team briefing & assign roles



BALANCE
Check balance & conditions



CLIMB
Execute climb sequence



SECURE
Secure connections & components



CERTIFY
Complete documentation



RETURN TO SERVICE
Test & return to normal operations

CRANE BALANCE

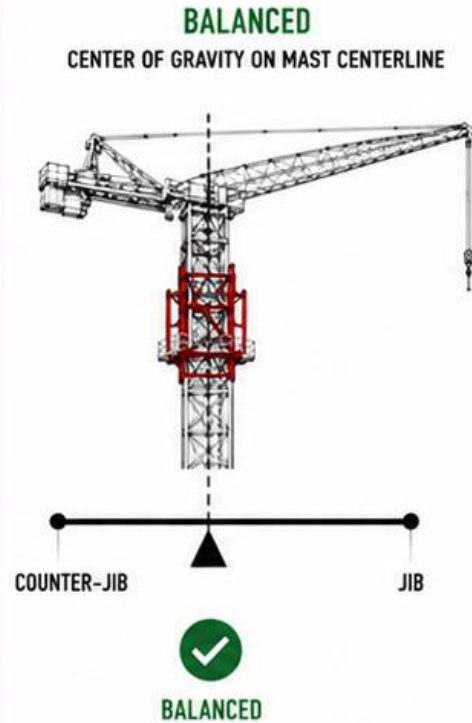
STAY BALANCED. STAY SAFE.

The crane must remain balanced at all times during climbing. Keep the center of gravity on the mast centerline as specified by the manufacturer.



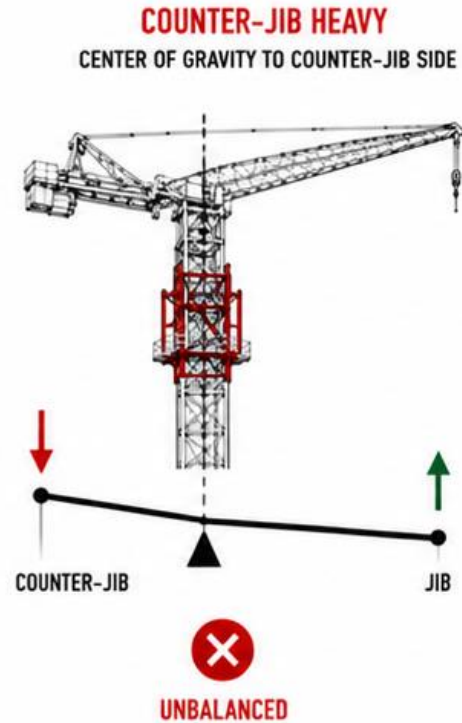
BALANCE IS CRITICAL

- Use only manufacturer-approved configurations and balancing loads.
- Do not climb if balance cannot be maintained.
- Recheck balance after each step, installation, or configuration change.
- An unbalanced crane increases stress on the structure and can lead to damage or tipping.

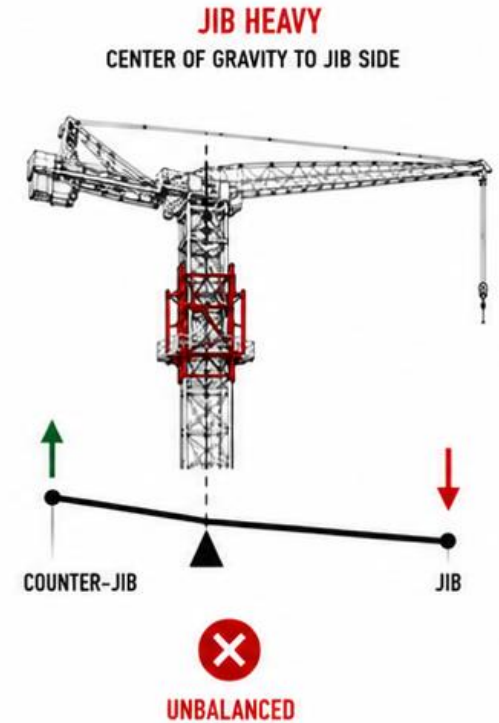


Within manufacturer's specified limits.

BALANCE DURING CLIMBING



Excess load on counter-jib.
Do not climb.



Excess load on jib.
Do not climb.

THE CLIMBING PROCESS



PLAN
Pre-climb planning



PREPARE
Set up & prepare



INSPECT
Inspect & verify



BRIEF
Team briefing & assign roles



BALANCE
Check balance & conditions



CLIMB
Execute climb sequence



SECURE
Secure connections & components



CERTIFY
Complete documentation



RETURN TO SERVICE
Test & return to normal operations



TOP CLIMBING – UNDERSTAND THE SEQUENCE

FOLLOW THE MANUFACTURER'S PROCEDURE. NEVER IMPROVISE.

The climbing sequence varies by crane model and manufacturer. You must follow the manufacturer's instructions exactly, maintain balance at all times, and use only trained, authorized personnel.



WHY SEQUENCE MATTERS

- Maintains structural integrity
- Prevents overload and tipping
- Ensures load is transferred safely
- Protects people, equipment and the crane

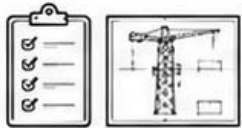


CONDITIONS THAT AFFECT CLIMBING

- Wind, freezing rain, temperature
- Snow, lightning, fog
- Stop the climb if conditions change
- Reassess and obtain authorization

1

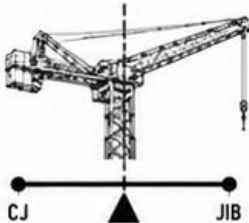
CONFIRM PLAN



- Review climbing plan and manufacturer's procedure
- Verify configuration and required mast height
- Confirm tools, parts, and personnel
- Obtain authorization to proceed

2

BALANCE CRANE



- Position trolley and hook as specified
- Configure counter-jib loads
- Center gravity on mast centerline as required
- Recheck balance

3

ENGAGE CLIMBING SYSTEM



- Inspect climbing unit, rams, hoses and controls
- Secure climbing dogs
- Connect hydraulic lines
- Test system operation

4

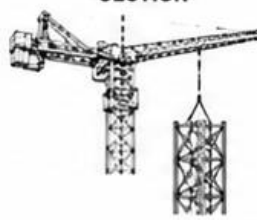
RAISE UPPER WORKS



- Operate climbing system to raise crane
- Maintain smooth, even lift
- Maintain balance
- Communication at all times

5

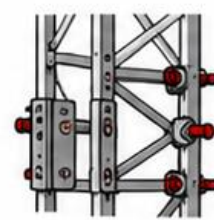
POSITION MAST SECTION



- Bring correct mast section into position
- Align holes and orientation
- Use tag lines – keep hands clear

6

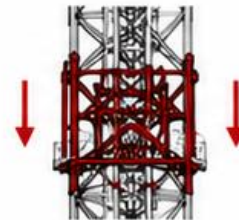
CONNECT & SECURE



- Install pins/bolts in the manufacturer's specified sequence
- Install all required hardware
- Tighten and secure
- Verify connections

7

TRANSFER LOAD



- Transfer crane load onto the new mast section
- Confirm climbing unit is supporting as required
- Follow manufacturer's procedure

8

VERIFY BEFORE CONTINUING



- Check all connections
- Verify balance
- Check climbing system position
- If all correct, proceed to next step of procedure



THE EXACT CLIMBING SEQUENCE IS MANUFACTURER-SPECIFIC – NEVER IMPROVISE OR SKIP A STEP.

If at any time the crane cannot be kept balanced, stop the climb and consult the manufacturer or your Climbing Supervisor.



REMEMBER

Only the Climbing Supervisor can authorize each step. Everyone else follows instructions.

THE CLIMBING PROCESS



PLAN
Pre-climb planning



PREPARE
Set up & prepare



INSPECT
Inspect & verify



BRIEF
Team briefing & assign roles



BALANCE
Check balance & conditions



CLIMB
Execute climb sequence



SECURE
Secure connections & components



CERTIFY
Complete documentation



RETURN TO SERVICE
Test & return to normal operations



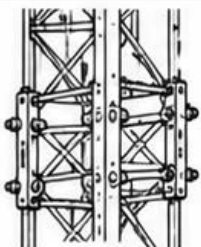
SECURE CONNECTIONS

EVERY CONNECTION. EVERY TIME.

All structural connections and climbing equipment must be secured in the manufacturer's specified sequence and to the required torque. Verify before proceeding.

WHAT TO SECURE & HOW TO VERIFY

1. STRUCTURAL CONNECTIONS



- Install all pins/bolts in the manufacturer's specified sequence
- Use correct hardware and locking devices
- Apply required torque
- Verify all connections are tight and secure

2. CLIMBING FRAME & TIE-IN CONNECTIONS



- Secure tie-in frames to mast
- Install retaining pins and locks
- Check for proper engagement of all locating devices
- Tighten all bolts to torque

3. CLIMBING UNIT & DOGS



- Verify dogs fully engaged
- Install and secure locking pins
- Check wear and condition
- Test by slight load or tug per manufacturer

4. HYDRAULIC RAMS & CONNECTIONS



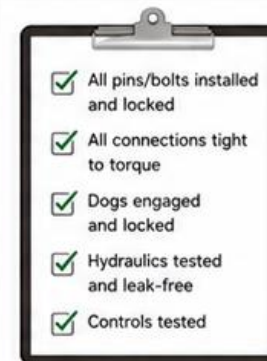
- Connect hoses to correct ports
- Check fittings for tightness and leaks
- Secure hose routing
- Pressure test system

5. ELECTRICAL & CONTROL CONNECTIONS



- Connect control plugs and cables
- Verify secure connections
- Check for damaged cables
- Test functions before climb continues

6. VERIFICATION



- Supervisor verifies and signs off
- Do not proceed until everything is verified



SECURE IS SAFE

- Improper or incomplete connections can lead to collapse or tipping.
- Use the correct pins, bolts, and hardware—never substitute.
- Follow torque values and manufacturer sequence.
- Recheck after load transfer.



CONNECTION ZONES (TYPICAL)

- 1 Upper tie-in / climbing frame
- 2 Mast section joint
- 3 Lower tie-in / climbing frame
- 4 Climbing unit connections
- 5 Hydraulic ram connections



STOP & CHECK

- If a connection does not align, do not force it.
- If any component is damaged or missing, stop the climb.
- Consult the manufacturer or your Climbing Supervisor.
- Never continue with doubt or pressure.

THE CLIMBING PROCESS



PLAN
Pre-climb planning



PREPARE
Set up & prepare



INSPECT
Inspect & verify



BRIEF
Team briefing & assign roles



BALANCE
Check balance & conditions



CLIMB
Execute climb sequence



SECURE
Secure connections & components



CERTIFY
Complete documentation



RETURN TO SERVICE
Test & return to normal operations



CERTIFICATION – CLIMB COMPLETE

DOCUMENT. VERIFY. AUTHORIZE RETURN TO SERVICE.

A climb is not complete until it has been verified, documented, and signed off by the authorized person.
 No load lifting or normal operations until certification is complete.



WHY CERTIFY?

- Confirms the climb was done safely and in the correct sequence
- Verifies all connections and systems
- Creates a legal and safety record
- Required by regulators and standards



CERTIFICATION CHECKLIST

1

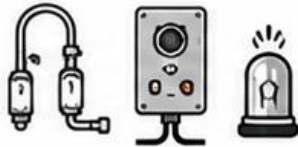
FINAL VERIFICATION



- All pins/bolts installed and locked
- All connections tight to torque
- Climbing unit and dogs properly engaged and secured
- Hydraulic rams and hoses secure
- No components missing or damaged

2

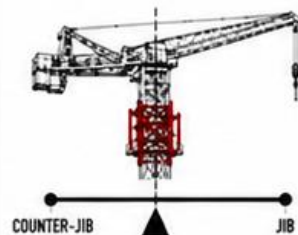
SYSTEMS CHECK



- Hydraulic system – normal operation
- Electrical and control systems – OK
- Anti-two-block and limit functions tested
- Safety devices and interlocks functional
- Lubrication points serviced

3

BALANCE CONFIRMED



- Crane balanced per manufacturer
- Configuration within allowable limits
- No side loading on climbing system

4

DOCUMENTATION



- Record climb details (date, time, height, configuration)
- Note any deviations and corrections
- Attach inspections and test results
- Retain records with crane file

5

AUTHORIZATION



- Climb verified by Climbing Supervisor
- Authorized person signs off
- Return to service approved
- Remove climb zone barricades when safe

6

RETURN TO SERVICE



- Resume normal operations
- Monitor crane performance
- Reconfirm balance after any further changes

REMEMBER

NO LOADS UNTIL CERTIFIED.

If anything is uncertain, stop and reassess.
 If required, consult the manufacturer or your Climbing Supervisor.



ALWAYS FOLLOW THE MANUFACTURER'S INSTRUCTIONS

THE CLIMBING PROCESS



PLAN
Pre-climb planning



PREPARE
Set up & prepare



INSPECT
Inspect & verify



BRIEF
Team briefing & assign roles



BALANCE
Check balance & conditions



CLIMB
Execute climb sequence



SECURE
Secure connections & components



CERTIFY
Complete documentation



RETURN TO SERVICE
Test & return to normal operations



RETURN TO SERVICE – CHECK & TEST

VERIFY PERFORMANCE BEFORE RESUMING OPERATIONS.

After each climb, the crane must be checked and tested to confirm it is operating correctly and safely before returning to service. All tests must be performed in accordance with the manufacturer's instructions and recorded.

POST-CLIMB CHECKS & TESTS

1

VISUAL INSPECTION



- Check all connections, pins, bolts and locks
- Climbing unit and dogs secure and stowed
- Hydraulic rams fully retracted and secured
- No loose parts or tools left on crane

2

LIMIT SWITCHES & LIMITS



- Test upper and lower limit switches
- Test hoist limits
- Test trolley limits
- Test boom/jib limits
- Test rotation limits

3

ANTI-TWO-BLOCK SYSTEM



- Test upper A2B
- Test lower A2B
- Verify audible/visual alarm and function
- Verify system resets correctly

4

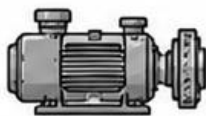
LOAD MOMENT INDICATOR (LMI)



- Power up and self-test
- Zero and calibrate as required
- Verify correct configuration (jib length, rope fall, etc.)
- Test rated capacity display

5

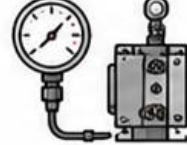
HOIST, TROLLEY & SLEW FUNCTIONS



- Hoist – up and down
- Trolley – in and out
- Slew – left and right
- Smooth operation, no abnormal noise or vibration
- Brakes hold load

6

HYDRAULIC / CLIMBING SYSTEM



- Check hydraulic oil level and for leaks
- Test climbing system pressure and release
- Verify hydraulic functions operate correctly

7

WEATHER & SAFETY DEVICES



- Test anemometer
- Test wind alarm
- Test obstruction lighting
- Verify all alarms and warnings

8

FINAL OPERATOR CHECK



- All tests completed and recorded
- Crane balanced
- No active alerts
- Crane ready for service



RECORD & REPORT

Record all post-climb checks and tests.

Report any defects immediately.

Keep records with crane file.

IF IN DOUBT, KEEP IT OUT!

SAFETY IS EVERYONE'S RESPONSIBILITY.





WHY TEST?

- Ensures the crane is safe to operate
- Verifies systems affected by the climb
- Confirms function of safety devices and limit systems
- Required by manufacturer, CSA Z248 and OH&S regulations



IMPORTANT

If any test fails, do not return the crane to service.

Correct the issue, retest, and re-verify.

Only the operator or authorized person can return the crane to service.

THE CLIMBING PROCESS



PLAN
Pre-climb planning



PREPARE
Set up & prepare



INSPECT
Inspect & verify



BRIEF
Team briefing & assign roles



BALANCE
Check balance & conditions



CLIMB
Execute climb sequence



SECURE
Secure connections & components



CERTIFY
Complete documentation



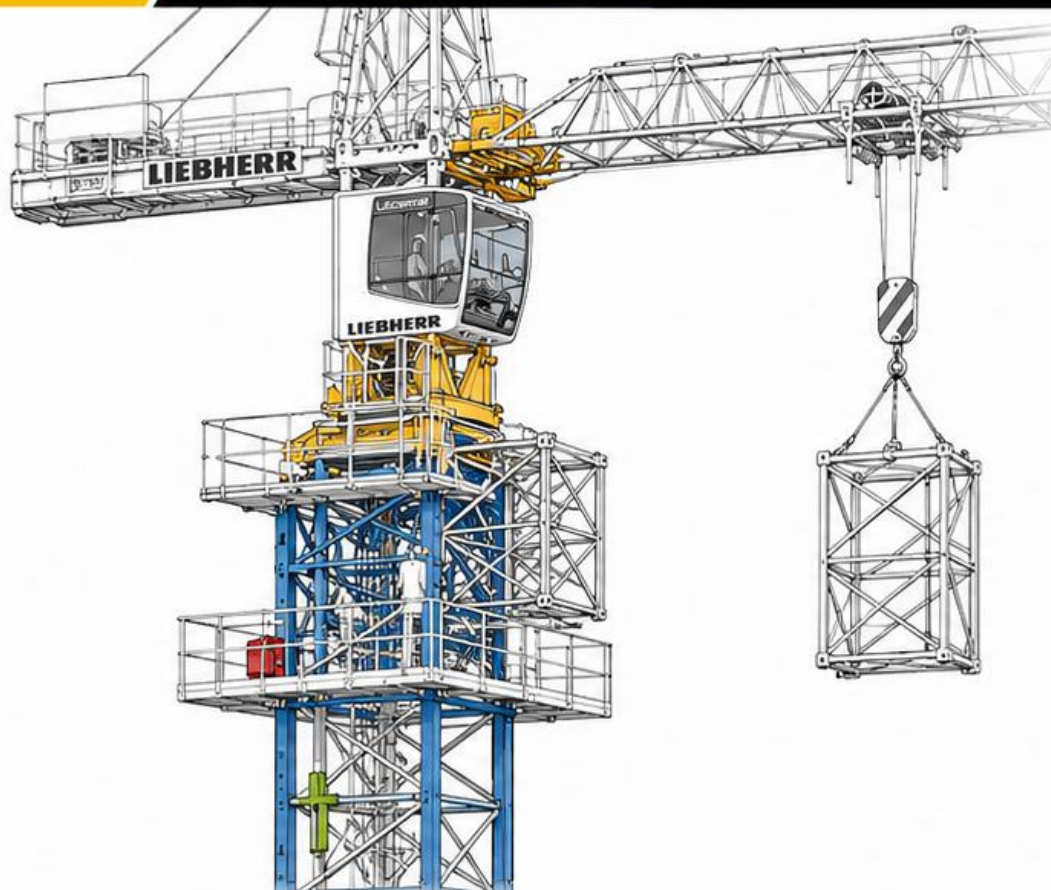
RETURN TO SERVICE
Test & return to normal operations



SYSTEM ORIENTATION

Liebherr's Fast Climbing System is designed to raise and support the upper crane so that new tower sections can be inserted and bolted into the tower.

The system alternates between hydraulic lifting (raising the upper crane) and positive structural support (support shoe and tower) to maintain safe, controlled climbing.



FAST CLIMBING SYSTEM COMPONENTS



GUIDE SECTION

Guides and stabilizes the upper crane during climbing.



HYDRAULIC RAM / SYSTEM

Provides the force that raises the upper crane.



CLIMBING CROSS-MEMBER

Engages the tower and provides the reaction point for the hydraulic ram.



SUPPORT SHOE

Supports the upper crane between hydraulic climbing movements.



TOWER SECTION

Inserted into the opening created by the climbing operation and bolted into the tower.

KEY BENEFITS



Safe & controlled climbing



Faster erection cycles



Modular & flexible tower solutions



Reduced site intervention



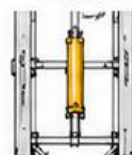
Proven worldwide performance

HOW THE SYSTEM WORKS ONE CLIMBING STAGE

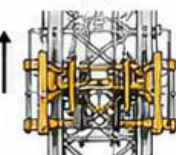
1 CROSS-MEMBER POSITIONED



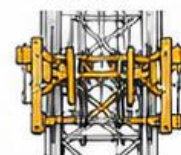
2 RAM EXTENDED



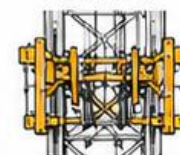
3 UPPER CRANE RAISED



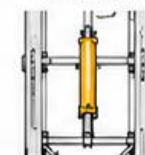
4 SUPPORT SHOE ENGAGED



5 LOAD TRANSFERRED TO SUPPORT



6 RAM RETRACTED



SOURCE (MANUFACTURER)

Liebherr – High-Top EC-H Technical Document
"Vorsprung durch Modularität" (Pages 19 – 22)



VIEW DOCUMENT (PDF)

https://www.liebherr.com/shared/media/construction-machinery/tower-cranes/brochures/top-slewing-cranes/high-top-ec-h/liebherr-vorsprung-durch-modularitaet_en.pdf



Always follow the specific manufacturer's instructions for your crane model.

LIEBHERR

HOW THE SYSTEM WORKS

One climbing stage allows the upper crane to be raised and supported so that the tower can be extended.

The system alternates between hydraulic lifting and positive structural support.



SAFETY BY DESIGN

- ✓ Monitored by limit switches and sensors
- ✓ Prevents incorrect operation
- ✓ Ensures correct positioning of supporting equipment
- ✓ Designed and tested by Liebherr to meet global standards

1 POSITION CROSS-MEMBER

The climbing cross-member is positioned on the tower at the climbing location.



2 EXTEND HYDRAULIC RAM

The hydraulic ram extends and pushes against the cross-member.



3 UPPER CRANE RISES

The hydraulic ram raises the upper crane safely upwards.



4 ENGAGE SUPPORT SHOE

The support shoe engages and takes load from the upper crane.



5 TRANSFER LOAD TO SUPPORT

Load is transferred from the hydraulic ram to the support shoe.



6 RETRACT HYDRAULIC RAM

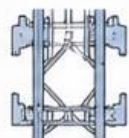
The hydraulic ram retracts fully, ready for the next climbing stage.



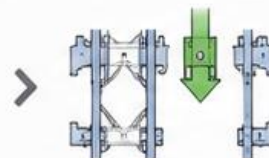
WHAT HAPPENS NEXT?



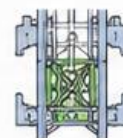
Two climbing stages create enough clearance to insert a new tower section (typically 2.5 m). The new section is inserted and bolted in place. The climbing cycle then continues.



Two climbing stages



Lower new section into opening



Bolt section in place



Tower extended – continue climbing



MANUFACTURER NOTE

Always follow the specific Liebherr operating instructions and site procedures for your crane model.



SOURCE (MANUFACTURER)

Liebherr – High-Top EC-H Technical Document
“Vorsprung durch Modularität” (Pages 19 – 22)



VIEW DOCUMENT (PDF)

https://www.liebherr.com/shared/media/construction-machinery/tower-cranes/brochures/top-slewing-cranes/high-top-ec-h/liebherr-vorsprung-durch-modularitaet_en.pdf



Always follow the specific manufacturer's instructions for your crane model.

LIEBHERR

TOWER SECTION INSERTION (FRONT ENTRY)

New tower sections are inserted from the front using the crane's own hook.

Two climbing stages create enough clearance. The section is then lifted on the front tray and inserted vertically into the opening beneath the turntable.

The upper crane (turntable and jib) remains supported above the opening at all times.

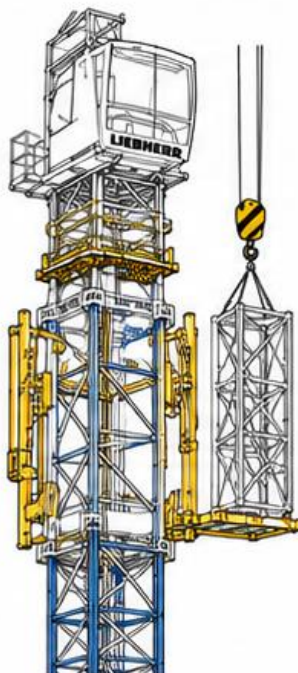


SAFETY BY DESIGN

- ✓ Monitored by limit switches and sensors
- ✓ Prevents incorrect operation
- ✓ Ensures correct positioning of supporting equipment
- ✓ Designed and tested by Liebherr to meet global standards

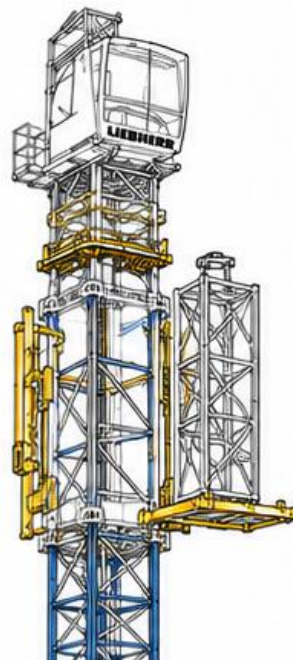
1 SECTION ON HOOK

New tower section is lifted by the crane's hook and brought to the front of the climber.



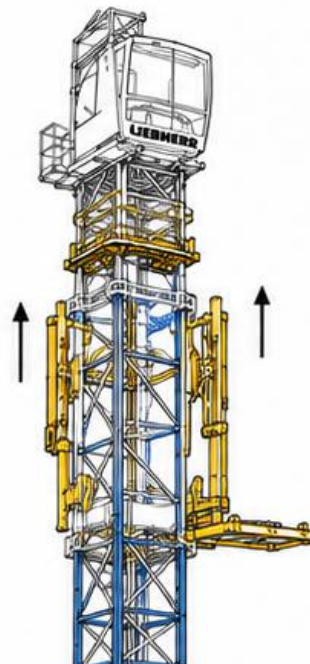
2 SECTION PLACED ON FRONT TRAY

The section is set down vertically on the front tray and secured for lifting.



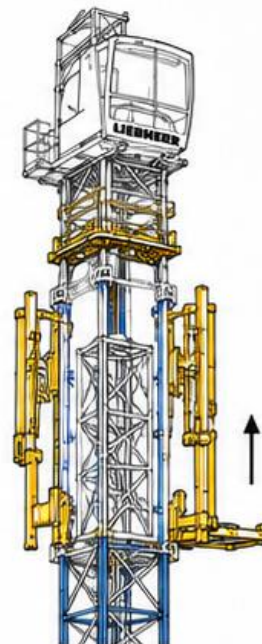
3 OPENING CREATED

The hydraulic ram extends and raises the upper crane and guide section, creating clearance for insertion.



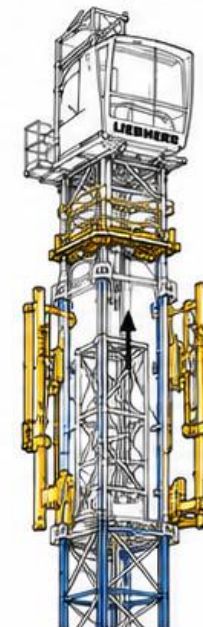
4 SECTION LIFTED INTO OPENING

The front tray lifts the section vertically and positions it in line with the opening.



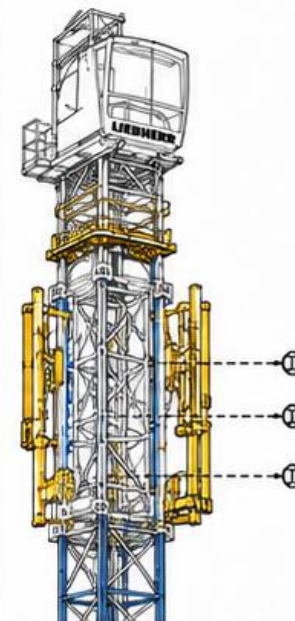
5 SECTION INSERTED VERTICALLY

The tray pushes the section straight into the opening. The climber is wider than the tower to allow the section to pass inside.



6 SECTION IN PLACE AND BOLTED

The section is fully seated in the tower and bolted through the connection points to the existing tower section.



KEY POINTS



Upper crane (turntable and jib) remains supported above the opening at all times.



Front tray carries the section and lifts it vertically into the opening.



Hydraulic ram raises the guide section and upper crane to create clearance.



Section is inserted vertically from the front, not turned or tilted.



All connection bolts must be tightened to the specified torque.



Verify alignment and secure connection before retracting the hydraulic ram.



SOURCE (MANUFACTURER)

Liebherr – High-Top EC-H Technical Document
"Vorsprung durch Modularität" (Pages 19 – 22)



VIEW DOCUMENT (PDF)

https://www.liebherr.com/shared/media/construction-machinery/tower-cranes/brochures/top-slewing-cranes/high-top-ec-h/liebherr-vorsprung-durch-modularitaet_en.pdf



Always follow the specific manufacturer's instructions for your crane model.

LIEBHERR

POTAIN SYSTEM OVERVIEW

Potain hydraulic climbing systems use a front mounted climbing cage with integrated hydraulic jacks to lift the upper crane.

Tower sections are inserted from the front, vertically, through the climbing cage. Once bolted, the load is transferred from the hydraulic jacks to the new tower section and the climber retracts for the next cycle.

SAFETY BY DESIGN

- ✓ Monitored by limit switches and pressure sensors
- ✓ Prevents incorrect operation
- ✓ Ensures correct positioning of supporting equipment
- ✓ Designed and tested by Manitowoc to meet global standards

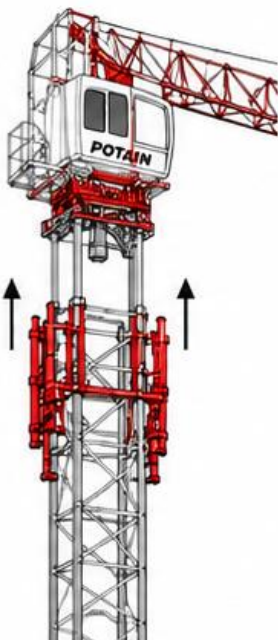
1 SYSTEM AT REST

Upper crane is supported by the tower. Climbing cage is attached to the tower.



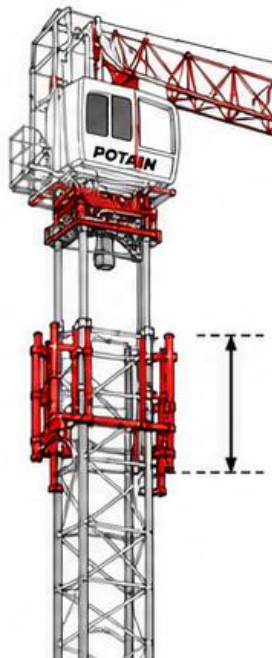
2 LIFT UPPER CRANE

Hydraulic jacks extend simultaneously to raise the climbing cage and the upper crane.



3 CREATE OPENING

Once the upper crane is raised, the opening is created for the new tower section.



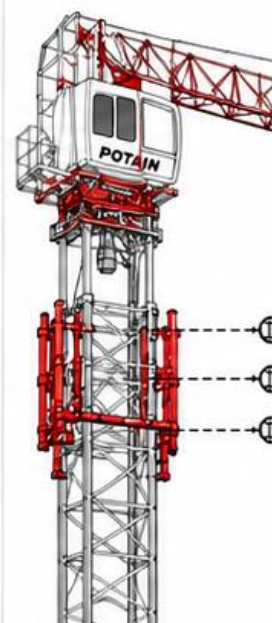
4 INSERT TOWER SECTION

New tower section is lifted by the crane hook and placed vertically in the front opening.



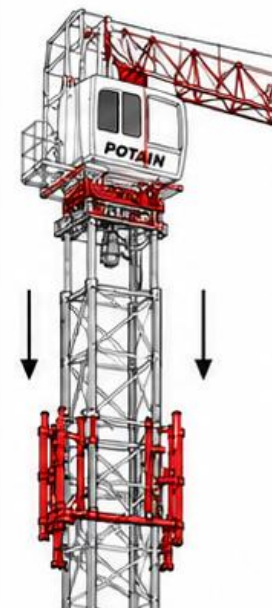
5 SECTION BOLTED

Section is fully seated and bolted through the connection points to the existing tower.



6 LOWER & RETRACT

Hydraulic jacks retract to transfer the load to the new section. The climbing cage lowers and clears the next lift position.



KEY POINTS



Front mounted climbing cage is wider than the tower to allow section insertion.



Hydraulic jacks lift the upper crane in unison to keep it level.



Section is lifted by the crane hook and inserted vertically from the front.



All bolts must be tightened to the specified torque before lowering.



Interlocks prevent lowering or retraction until load transfer is complete.



Always follow the manufacturer's instructions for your crane model.



SOURCE (MANUFACTURER)

Manitowoc – Potain Hydraulic Climbing Systems
Technical Documentation and Operator Manual



VIEW DOCUMENT (PDF)

<https://www.manitowoc.com/-/media/core/products/potain/documents/potain-hydraulic-climbing-system-manual.pdf>



Always follow the specific manufacturer's instructions for your crane model.

POTAIN
by **Manitowoc**

PAWL ENGAGEMENT & LOAD TRANSFER

Once the new section is inserted and bolted, the climbing cage lowers until the pawls engage the next set of tower lugs.

The load is then transferred from the hydraulic jacks to the pawls through the climbing cage.

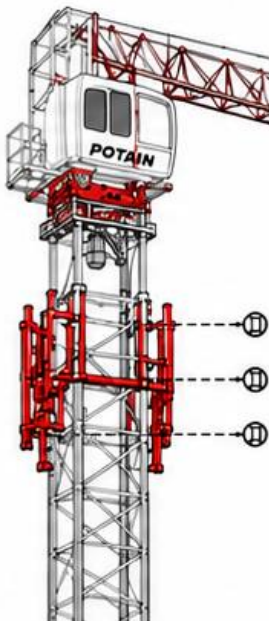
After verification, the jacks are released and the system is ready for the next climbing cycle.

SAFETY BY DESIGN

- ✓ Monitored by load cells and limit switches
- ✓ Positive pawl engagement verification
- ✓ Prevents accidental lowering
- ✓ Designed and tested by Manitowoc to meet global standards

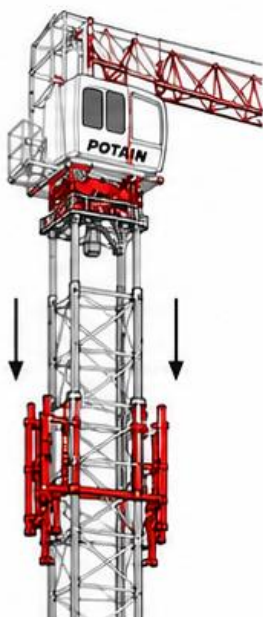
1 SECTION IN PLACE & BOLTED

New tower section is fully inserted and bolted to the existing tower.



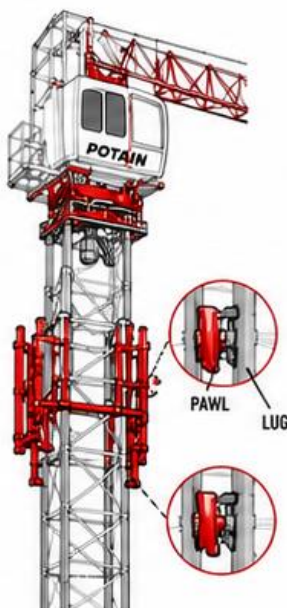
2 LOWER CLIMBING CAGE

Hydraulic jacks retract together to lower the climbing cage toward the next set of lugs.



3 PAWLS ALIGN WITH LUGS

Climbing cage continues lowering until pawls are aligned with the next set of tower lugs.



4 PAWLS ENGAGE

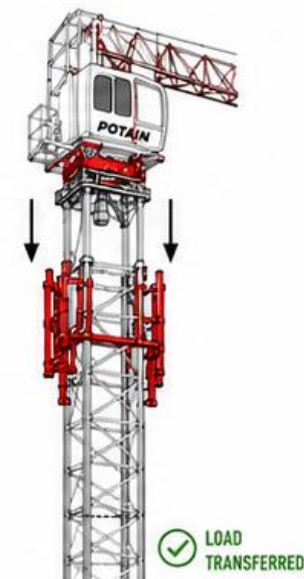
Pawls rotate and engage the lugs. Engagement is confirmed by position sensors.



5 LOAD TRANSFER

Hydraulic jacks continue to retract slightly to fully transfer the load from jacks to pawls.

Load cells verify zero load on jacks.



6 JACKS RELEASED

With load on pawls, hydraulic jacks are fully retracted.

System is locked and ready for the next climbing cycle.



KEY POINTS



Pawls provide the primary structural support for the upper crane.



Load cells confirm zero load on jacks before release.



Interlocks prevent jack release unless all pawls are fully engaged.



Sensors and limit switches ensure safe operation throughout the entire climbing sequence.



Hydraulic jacks are used only for lifting/lowering, not for supporting the load.



SOURCE (MANUFACTURER)

Manitowoc – Potain Hydraulic Climbing Systems
Technical Documentation and Operator Manual



VIEW DOCUMENT (PDF)

<https://www.manitowoc.com/-/media/core/products/potain/documents/potain-hydraulic-climbing-system-manual.pdf>



Always follow the specific manufacturer's instructions for your crane model.

POTAIN

by Manitowoc

TELESCOPING CLIMBING SEQUENCE

The Potain system uses a telescoping lower climbing cage. As the cage lowers, it compactly folds for transport and then extends to the full climbing height ready for the next lifting phase.

This allows efficient cycling with minimal crane downtime.

SAFETY BY DESIGN

- ✓ Monitored by position sensors throughout sequence
- ✓ Interlocks prevent lifting until cage is fully extended and locked
- ✓ Load cells verify zero load on jacks before release
- ✓ Designed and tested by Manitowoc to meet global standards

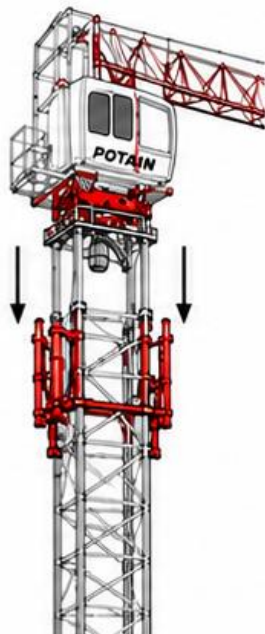
1 PAWLS ENGAGED

Pawls are engaged on the next set of tower lugs. Load is fully transferred from jacks to pawls.



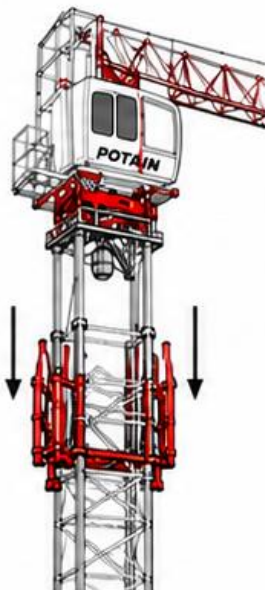
2 JACKS RELEASED

Hydraulic pressure is released and jacks are free from load.



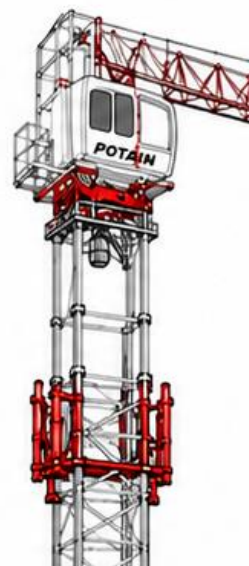
3 LOWER CAGE RETRACTS

The hydraulic cylinders retract together, lowering the climbing cage. The cage telescope collapses for transport.



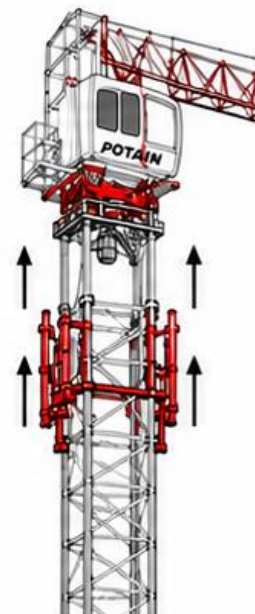
4 CAGE AT BOTTOM

Lower cage reaches the bottom position. Cage is compact for minimum wind exposure and clearance.



5 CAGE EXTENDS

Hydraulic cylinders extend, raising and expanding the climbing cage to the full climbing height.



6 READY FOR NEXT LIFT

Climbing cage is fully extended and locked in position. System is ready to create opening for the next tower section.



KEY POINTS



Cage lowers uniformly on all cylinders for stability and alignment.



Cage telescopes in as it lowers for compact transport position.



Interlocks ensure cage is fully extended and locked before the next lift is allowed.



LOAD

Zero load on jacks is verified before they are released.



Compact position reduces wind loading during lowering and travel.



SOURCE (MANUFACTURER)

Manitowoc – Potain Hydraulic Climbing Systems
Technical Documentation and Operator Manual



VIEW DOCUMENT (PDF)

<https://www.manitowoc.com/-/media/core/products/potain/documents/potain-hydraulic-climbing-system-manual.pdf>



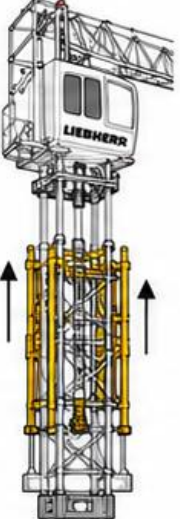
Always follow the specific manufacturer's instructions for your crane model.

POTAIN
by Manitowoc

COMPARISON	LIEBHERR FAST CLIMBING SYSTEM	POTAIN HYDRAULIC CLIMBING SYSTEM
 SYSTEM TYPE	Single-ram external guide section system.	Telescoping cage system with integrated hydraulic jacks.
 SECTION INSERTION	Front entry on work tray. Tower section inserted vertically within guide section.	Front entry. Tower section inserted vertically through front opening of cage.
 LOAD TRANSFER	Hydraulic ram lifts upper crane. Load transferred straight down through guide section to support shoe.	Pawls engage tower lugs. Load transferred from jacks to pawls. Jacks retract.
 LIFTING METHOD	One (1) central hydraulic ram.	Multiple hydraulic jacks (typically 4 at the corners).
 CLIMBING CAGE GEOMETRY	Guide section wider than tower. Encloses tower section on all sides.	Telescoping lower cage. Folds for transport, extends to full height.
 CYCLE TIME	Generally faster (fewer steps to complete a cycle).	Slightly longer (additional steps for pawl engagement and jack retraction).
 MAINTENANCE	Fewer moving parts (central ram system).	More components (jacks, pawls, lugs, telescoping cage).
 WIND SENSITIVITY	Less wind exposed (compact guide around tower).	More wind exposed (larger cage at full height).

LIEBHERR FAST CLIMBING SYSTEM

1 LIFT UPPER CRANE
Hydraulic ram lifts the upper crane.



2 INSERT SECTION
New tower section is inserted vertically within guide section.



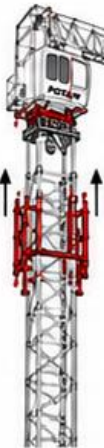
SECTION FULLY CONTAINED WITHIN GUIDE SECTION

3 LOAD TRANSFER
Load is transferred straight down through guide section to the support shoe.

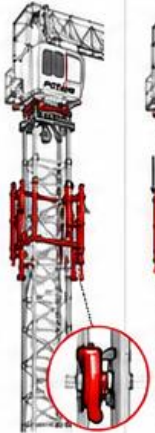


POTAIN HYDRAULIC CLIMBING SYSTEM

1 JACKS EXTEND
Jacks lift the climbing cage and upper crane.



2 PAWLS ENGAGE
Pawls rotate and engage tower lugs. Engagement confirmed by sensors.



3 JACKS RETRACT
Jacks retract and transfer load from jacks to pawls.



4 CAGE LOWERS
Lower cage telescopes down to the bottom position.



5 CAGE EXTENDS
Jacks extend and raise the cage to full climbing height.



6 READY FOR NEXT LIFT
System is locked and ready for next tower section.



READY FOR NEXT CYCLE

KEY TAKEAWAYS



Both systems are proven, safe and efficient when used as designed.



Liebherr uses a single ram and guide section for a fast, direct load transfer.



Potain uses pawls and multiple jacks with a telescoping cage for greater versatility.



Consider wind, site conditions and maintenance when selecting or operating.



Always follow the specific manufacturer's instructions for your crane model.



SOURCES (MANUFACTURERS)

Liebherr – High-Top EC-H Technical Document "Vorsprung durch Modularität" (Pages 19 – 22)

Manitowoc – Potain Hydraulic Climbing Systems Technical Documentation and Operator Manual



VIEW DOCUMENTS (PDF)

www.liebherr.com
www.manitowoc.com



Always follow the specific manufacturer's instructions for your crane model.

RED SEAL KNOWLEDGE OVERVIEW

TOWER CRANE CLIMBING & RECONFIGURATION (TASK C-7)



CURRENT RED SEAL
OCCUPATIONAL STANDARD (RSOS)
2023 EDITION

Interprovincial exam implemented
June 2025

8

8 QUESTIONS
out of 100 on the
Red Seal Exam



TESTS YOUR
Knowledge
Procedures
Application
Critical Thinking



APPLY YOUR KNOWLEDGE
to the crane and climbing
system in use.

There is no single universal
manufacturer sequence.

TASK C-7 – CLIMBING & RECONFIGURATION



Participates in bottom
climbing procedures.



Participates in top
climbing procedures.



Participates in crane
reconfiguration procedures.

REMEMBER

- ✓ Follow the **SPECIFIC** manufacturer's climbing procedure.
- ✓ Understand **WHY** the sequence, balancing and procedures are important.
- ✓ Recognize hazards, weather effects and safe work practices.
- ✓ Ensure plan, equipment, standards and approvals are in place before climbing.

REFERENCES



Red Seal Occupational Standard (RSOS)
Tower Crane Operator – 2023 Edition
red-seal.ca/rsos
(Direct PDF: [rsos-eng.pdf](#))



Red Seal Exam Information
Tower Crane Operator
red-seal.ca/exam-information
(Includes sample questions)



Red Seal Exam Weightings
Tower Crane Operator
red-seal.ca/exam-weightings



SkilledTradesBC
Tower Crane Operator
skilledtradesbc.ca/tower-crane-operator



Crane Certification Organization (CCO)
Operator Certification Program
coc.ca

STANDARDS & REGULATIONS



CSA Z248
Tower Cranes



OHS Regulation
(Part 14 – Crane Safety)



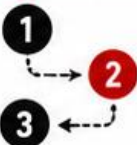
Provincial / Territorial
OH&S Regulations
(As Applicable)

RED SEAL KNOWLEDGE REQUIREMENTS

TASK C-7.02 – TOP-CLIMBING PROCEDURES



The Red Seal Occupational Standard (RSOS) requires the Tower Crane Operator to demonstrate knowledge of the following for **C-7.02 Top-Climbing Procedures**. These are examinable topics on the Red Seal exam (**8 questions** total for Task C-7).

1 TOOLS & EQUIPMENT AND THEIR USE  - Identify tools and equipment used for top climbing. - Know the purpose of each item. - Know the procedures for their safe and effective use.	2 HAZARDS & SAFE WORK PRACTICES  - Recognize hazards associated with top climbing. - Apply safe work practices to control risks. - Protect personnel, equipment and the public.	3 PRE-CLIMBING ACTIVITIES  - Plan and prepare for the climb. - Confirm supports, power, grounding and shoring. - Verify building material strength (engineering approval).
4 PROCEDURES TO PERFORM TOP CLIMBING  - Understand the complete top-climbing procedure for the system being used. - Know each step and the purpose of each step. - Follow the manufacturer's procedure.	5 IMPORTANCE OF CLIMBING SEQUENCE  - Understand why steps must be performed in the correct order. - Know the consequences of an incorrect sequence. - Maintain control and stability throughout the sequence.	6 PROCEDURES TO BALANCE THE CRANE  - Understand how and why the crane must be balanced. - Know the procedures for maintaining balance during climbing. - Follow manufacturer requirements for balance.
7 EFFECTS OF WEATHER CONDITIONS  - Identify weather conditions that can affect climbing. - Assess wind, freezing rain, temperature, snow, lightning and fog. - Know when to delay, stop or resume climbing.	8 MANUFACTURER-SPECIFIC TOP-CLIMBING TRAINING  - Understand and follow the specific manufacturer's top-climbing training. - Use the correct configuration, components and procedures for that crane model.	9 STANDARDS & REGULATIONS  - Know the applicable standards, codes and regulations. - Follow CSA Z248 and other relevant standards. - Comply with jurisdictional OH&S requirements.



RED SEAL MESSAGE:

You must understand the **what, why and how** of top-climbing operations. There is no single universal sequence – follow the **specific manufacturer's procedure**.



Expect questions that test your knowledge, application and critical thinking.

TASK C-7 OVERVIEW (8 QUESTIONS)

- C-7.01 Bottom-Climbing Procedures
- C-7.02 Top-Climbing Procedures**
- C-7.03 Reconfiguration Procedures



OFFICIAL
REFERENCES



Red Seal Occupational Standard (RSOS)
Tower Crane Operator – 2023 Edition
red-seal.ca/rsos
(Direct PDF: [rsos-eng.pdf](#))



Red Seal Exam Information
Tower Crane Operator
red-seal.ca/exam-information
(Includes sample questions)



Red Seal Exam Weightings
Tower Crane Operator
red-seal.ca/exam-weightings



SkilledTradesBC
Tower Crane Operator
skilledtradesbc.ca/tower-crane-operator



Crane Certification Organization (CCO)
Operator Certification Program
cco.ca



The Red Seal Occupational Standard (RSOS) requires the Tower Crane Operator to demonstrate knowledge of **pre-climbing activities** and **hazards and safe work practices** associated with top-climbing operations. (C-7.02 Top-Climbing Procedures)

PRE-CLIMBING ACTIVITIES

(Knowledge Requirements)



1 PLAN & CONFER

Plan the climb and confer with the climbing team. Confirm the climbing plan is understood by all.



2 SUPPORT BEAMS

Prepare and install support beams as required by the manufacturer.



3 POWER CABLE

Ensure adequate power cable is available for the climb.



4 GROUNDING

Ensure proper grounding of the crane and climbing system.



5 SHORING

Ensure all required shoring is installed and secure.



6 BUILDING STRENGTH

Confirm that building material strength is adequate for climbing (engineering approval as required).



COMPLETE ALL PRE-CLIMBING ACTIVITIES BEFORE CLIMBING BEGINS

HAZARDS & SAFE WORK PRACTICES

(Knowledge Requirements)



1 CHANGING WEATHER

Wind, rain, snow, fog, lightning and temperature changes can affect climbing operations.



2 PERSONNEL IN AREA

Workers in the area can be struck by falling objects or caught in moving components.



3 OBSTRUCTIONS

Power lines, structures and other obstacles can create contact or interference hazards.



4 TRIPPING / FALLING

Slip, trip and fall hazards exist on the crane, mast, platform and climbing system.



5 ELECTROCUTION

Contact with energized power lines or exposed electrical components.



6 PINCH / CRUSH POINTS

Moving parts of the climbing system can cause serious pinch or crush injuries.



7 TIGHT CLEARANCES

Limited space can cause struck-by or caught-between hazards.



8 MECHANICAL FAILURE

Failure of climbing components, hydraulics or structure can lead to collapse or drop.



9 POOR VISIBILITY

Reduced lighting or visibility increases the risk of mistakes and incidents.



APPLY SAFE WORK PRACTICES

Follow the manufacturer's instructions, use appropriate PPE and never bypass safety devices.



REMEMBER

Recognize and control these hazards and complete all pre-climbing activities before starting the climb, as required by the RSOS.



RED SEAL EXAM TIP

You may be asked what must be done before climbing begins or which hazard requires the climb to stop.

TASK C-7 OVERVIEW (8 QUESTIONS)

- C-7.01 Bottom-Climbing Procedures
- **C-7.02 Top-Climbing Procedures (4–5 questions)**
- C-7.03 Reconfiguration Procedures



QUESTIONS ON EXAM
(Task C-7)



OFFICIAL
REFERENCES



Red Seal Occupational Standard (RSOS)
Tower Crane Operator – 2023 Edition
red-seal.ca/rsos
(Direct PDF: [rsos-eng.pdf](#))



Red Seal Exam Information
Tower Crane Operator
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(Includes sample questions)



Red Seal Exam Weightings
Tower Crane Operator
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SkilledTradesBC
Tower Crane Operator
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Crane Certification Organization (CCO)
Operator Certification Program
cco.ca



Always follow the manufacturer's instructions and applicable OH&S regulations for your crane model and jurisdiction.

SAFE PLANNING • SAFE SETUP • SAFE CLIMB • SAFE OPERATION



i The Red Seal Occupational Standard (RSOS) requires the Tower Crane Operator to demonstrate knowledge of **procedures to balance the crane** and the **effects of weather conditions** on top-climbing operations. (C-7.02 Top-Climbing Procedures)



WIND – TOP CLIMBING



MANUFACTURER'S CLIMBING LIMIT

There is no universal WorkSafeBC climbing wind limit.

Follow the manufacturer's specified wind limit for the particular crane and climbing procedure.



50 km/h ≠ CLIMBING LIMIT

50 km/h is WorkSafeBC's general maximum for **NORMAL TOWER CRANE OPERATION** when the manufacturer has not specified a maximum operating wind speed. It is **NOT** a climbing limit.



EXAMPLE – MANUFACTURER SPECIFICATION

If the climbing manual states maximum climbing wind = 37 km/h → **37 km/h** is the maximum for that climb.



TEMPERATURE – STOP LIMIT

-18°C (0°F)

BC STOP LIMIT

Tower crane operations must stop when the ambient temperature is below -18°C (0°F) unless otherwise specified by the manufacturer or a professional engineer.



WHAT THIS MEANS

- Climbing, operating, erecting or dismantling must stop below -18°C (0°F).
- Resume only when temperature is above the manufacturer's or P.Eng. approved lower limit.



WorkSafeBC OHSR 14.93:

Tower crane operations must stop when the temperature is below -18°C (0°F) unless otherwise specified by the manufacturer or a professional engineer.



LIGHTNING – STOP / SEEK SHELTER

Follow Environment and Climate Change Canada lightning safety guidance.



**IF YOU CAN HEAR THUNDER
SEEK SHELTER
IMMEDIATELY**



**WAIT AT LEAST
30 MINUTES**

**AFTER THE LAST THUNDER
BEFORE RETURNING OUTDOORS.**



Tower cranes and their wire ropes are large conductive structures. Lightning can travel through the crane and cause serious injury or death.



Monitor local conditions and lightning activity:
WeatherCAN / Canada's Weather Information Map
Lightning Layer: www.canada.ca/lightning-map

OTHER WEATHER CONDITIONS THAT CAN AFFECT TOP-CLIMBING OPERATIONS



FREEZING RAIN

Ice accumulation on components can reduce capacity, prevent movement and create slipping hazards.



SNOW

Snow build-up can add load, reduce visibility and create slippery surfaces.



FOG

Fog reduces visibility which increases the risk of collisions and miscommunication.



VISIBILITY

Poor visibility makes it harder to see hazards, hand signals and critical clearances.



GUSTS / CHANGES

Sudden wind gusts or rapid changes in wind speed can overload the crane.



SAFETY REMINDERS

- Use a calibrated anemometer at climbing height.
- Monitor weather continuously throughout the climb.
- Communicate any changing conditions immediately.
- Stop the climb if limits are approached or exceeded.
- Follow manufacturer instructions and all applicable regulations.



REFERENCES



WorkSafeBC OHS Regulation
Part 14 – Cranes and Hoists
Sections 14.73.2 • 14.92(4)(6) • 14.93
worksafebc.com



WorkSafeBC – Tower Crane
Guidelines (Part 14)
Understanding wind and weather
effects and limitations.
worksafebc.com



BC Crane Safety
Top Climbing a Tower Crane
Follow manufacturer's instructions
and safe work practices.
bccranesafety.ca



Environment and Climate Change Canada
Lightning Safety
canada.ca/lightning-safety
Lightning Map
canada.ca/lightning-map



Crane Certification Organization (CCO)
Tower Crane Operator
Certification Program
cco.ca



Always follow the manufacturer's instructions and all applicable OH&S regulations and jurisdictional requirements.



SAFE PLANNING



SAFE SETUP



SAFE CLIMB



SAFE OPERATION

RED SEAL KNOWLEDGE REQUIREMENTS

TASK C-7.02 – TOP-CLIMBING: SEQUENCE & EXAM THINKING



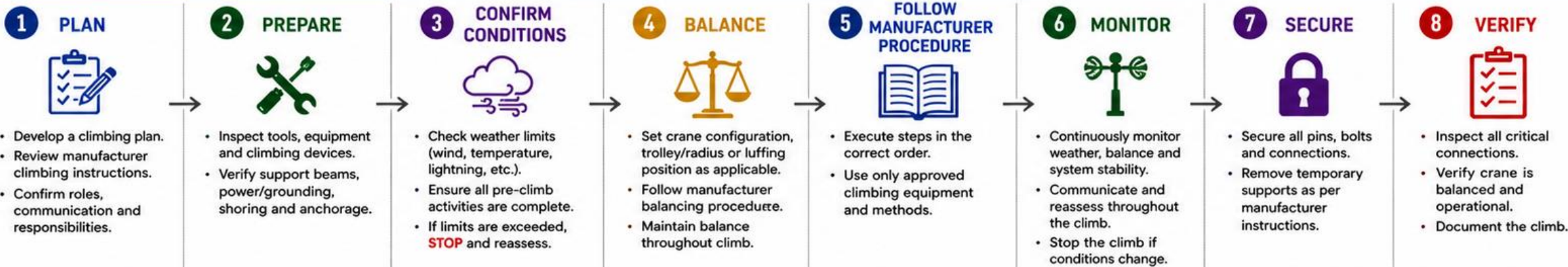
The Red Seal exam **does NOT** provide one universal top-climbing sequence. You must understand the importance of climbing sequence and be able to apply the correct steps and decision making based on manufacturer instructions, regulations and safe work practices.

8

RED SEAL EXAM WEIGHTING

8 questions on this task (Task C-7).

A LOGICAL TOP-CLIMBING PROCESS



WHY CLIMBING SEQUENCE MATTERS



- Correct sequence maintains crane balance and prevents tipping or structural failure.
- Out-of-sequence actions can overload components, create instability, or cause serious injury or death.
- Each manufacturer has specific requirements – follow them exactly.
- Sequence, balance and monitoring are tested on the Red Seal exam.

EXAMPLE RED SEAL EXAM QUESTIONS (Task C-7)

- | | | |
|---|---|---|
| Q | What must be confirmed before top-climbing begins? | → All pre-climbing activities, weather conditions, supports, balance and manufacturer requirements. |
| Q | Which of the following conditions requires the climb to stop? | → Wind above the manufacturer's climbing limit, temperature below -18°C, lightning within the unsafe range, or instability. |
| Q | Why must crane balance be maintained during the climbing operation? | → To prevent overturning, structural overload and component failure during lifting, setting or reconfiguration. |

EXAM THINKING – MAKE THE RIGHT CALL



Identify the hazard or condition.



Apply regulations, manufacturer requirements and safe work practices.



Decide the safest action. Protect people, equipment and the public.

WHEN IN DOUBT – STOP, REASSESS & FOLLOW THE MANUFACTURER'S INSTRUCTIONS.



OFFICIAL REFERENCES



Red Seal Occupational Standard (RSOS)
Tower Crane Operator – 2023 Edition
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Operator Certification Program
cco.ca



Always follow the manufacturer's instructions and applicable OH&S regulations and jurisdictional requirements.

SAFE PLANNING

SAFE SETUP



SAFE CLIMB

SAFE OPERATION

ANSWER SECTION

WITH EXPLANATIONS & REFERENCES

Detailed answers with the reasoning and references you need to master the standard.



CORRECT ANSWER

The best answer is identified.



EXPLANATION

Why the correct answer is correct.



REFERENCE

Exact source and section for further study.



EXAM FOCUS

Red Seal thinking, decision making and safety focus.



REAL WORLD

Apply knowledge on the job. Work safe.

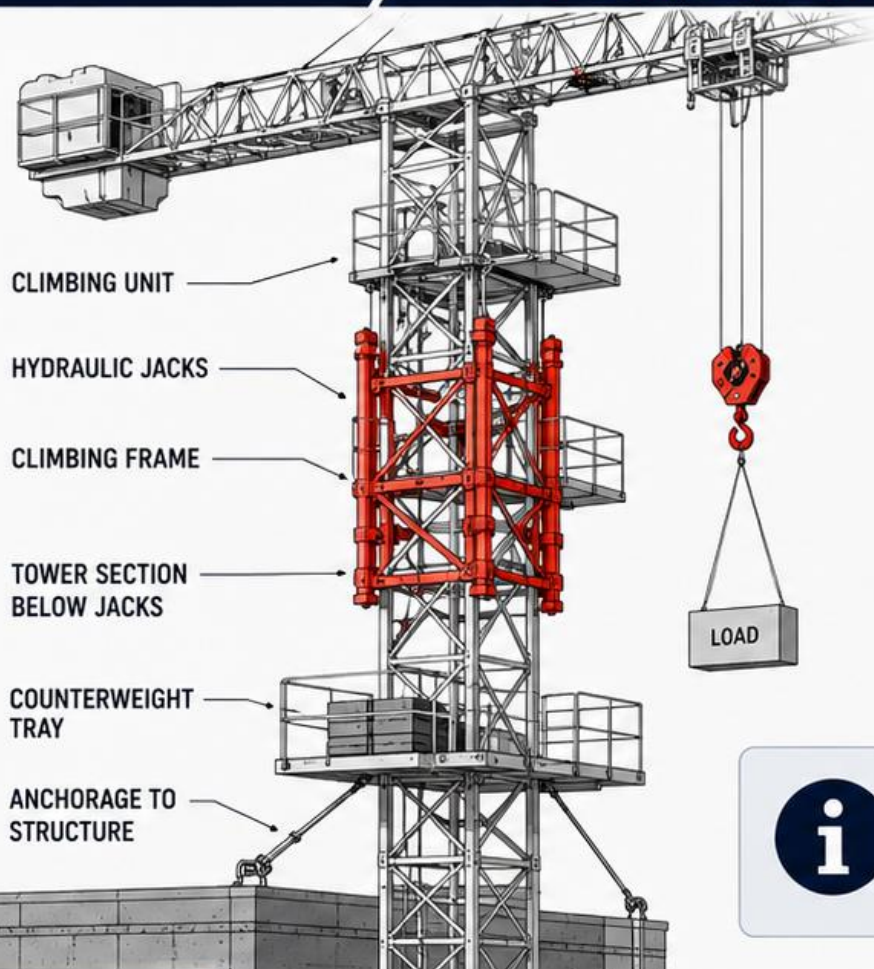


These answers are based on the Red Seal Occupational Standard (RSOS) for Tower Crane Operator – Task C-7.02: Top-Climbing – Pre-Climbing & Hazards.



IMPORTANT

Always follow the manufacturer's instructions, applicable regulations and safe work practices. When in doubt – **STOP**, reassess and seek guidance.



KEY REFERENCES



WorkSafeBC
OHS Regulation – Part 14
Cranes and Hoists
Sections 14.73 – 14.93



Red Seal Occupational Standard (RSOS)
Tower Crane Operator – 2023 Edition
Task C-7.02
Top-Climbing – Pre-Climbing & Hazards.



Manufacturer
Climbing Manuals
Follow specific crane
climbing procedures.



Environment Canada
Weather & Lightning
Monitor conditions and
follow safety guidance.



CSA Z248
Tower Cranes
Safety requirements
and best practices.





QUESTION 1

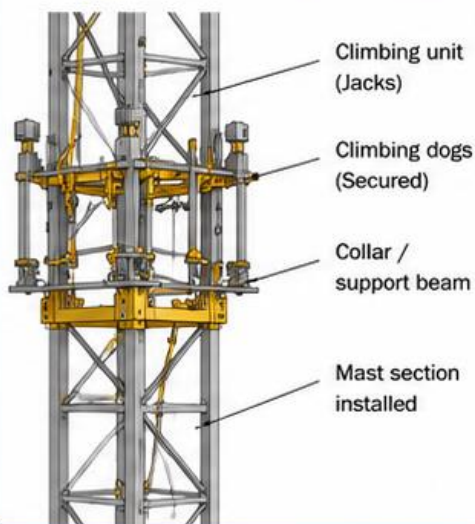
In British Columbia, erection, climbing, repositioning and dismantling of a tower crane must be performed:



- A. By any certified tower crane operator
- B. By qualified persons in accordance with the manufacturer's instructions, or professional engineer's instructions where applicable**
- C. Only by a manufacturer's representative
- D. By any worker supervised by the crane operator



TOP CLIMBING SYSTEM (EXAMPLE)



CORRECT ANSWER: B

WorkSafeBC OHS Regulation requires that crane erection, climbing, repositioning and dismantling be performed by qualified persons in accordance with the crane manufacturer's instructions.

If the installation varies from the manufacturer's instructions, written instructions from a professional engineer are required.

WHY THE OTHER CHOICES ARE WRONG

- A. Operator certification (e.g., Level B or D) authorizes a person to operate the crane, but does not qualify them to perform erection, climbing, repositioning or dismantling.
- C. The regulation allows qualified persons to perform the work; it is not limited to the manufacturer's representatives.
- D. General supervision by the operator does not meet the requirement for the work to be performed by qualified persons following the applicable instructions.

REFERENCE BASIS



WORKSAFEBC REGULATION

- OHS Regulation, Part 14 – Cranes and Hoists
14.73.2(1)
Erection, climbing, repositioning and dismantling of a tower crane must be performed by qualified persons in accordance with the crane manufacturer's instructions, or, if the installation varies from the crane manufacturer's instructions, in accordance with written instructions from a professional engineer.



RED SEAL RSOS (2023)

- C-7.00 Climbing and Reconfiguration
General performance outcome for Task C-7.



BC CRANE SAFETY OPERATOR STANDARD

- Section 6 – Top Climbing
Operators must follow manufacturer's procedures and use qualified personnel for climbing activities.



SKILLEDTRADESBC PROGRAM

- Tower Crane Operator Program
Training aligned to Red Seal RSOS and WorkSafeBC regulatory requirements.



KEY POINT: Only qualified persons, following the applicable manufacturer's instructions or professional engineer's instructions where required, may perform erection, climbing, repositioning and dismantling of a tower crane in British Columbia.





TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 2 – ANSWER

QUESTION 2

Who must direct a tower crane climbing operation?

- A. The prime contractor
- B. The crane operator
- ✓ C. A qualified supervisor
- D. The site superintendent

QUALIFIED SUPERVISOR

- ✓ Directs the climbing operation
- ✓ Coordinates the crew and operator
- ✓ Ensures the procedure follows manufacturer's instructions
- ✓ Has the training, knowledge and experience for the particular crane and activity



TYPICAL TOP CLIMB SEQUENCE (SIMPLIFIED)



CORRECT ANSWER: C A QUALIFIED SUPERVISOR

WorkSafeBC OHS Regulation **14.73.2(2)** requires that erection, climbing, repositioning and dismantling of a tower crane be performed under the direction of a qualified supervisor.

The qualified supervisor must be present and direct the operation, ensuring the work is done in accordance with the applicable manufacturer's instructions (or professional engineer's instructions where applicable).



THE SUPERVISOR
IS RESPONSIBLE
FOR THE SAFE
EXECUTION OF
THE CLIMB.

WHY THE OTHER CHOICES ARE WRONG

- A. The prime contractor has overall project responsibilities but is not required to direct the technical climbing operation.
- B. The crane operator operates the controls but does not direct the climbing operation.
- D. The site superintendent manages the site but is not required by regulation to direct the climbing operation unless they are also the qualified supervisor.

REFERENCE BASIS

 WORKSAFEBC REGULATION	OHS Regulation, Part 14 – Cranes and Hoists 14.73.2(2) <i>Erection, climbing, repositioning and dismantling of a tower crane must be performed under the direction of a qualified supervisor.</i>
 RED SEAL RSOS (2023)	- C-7.00 Climbing and Reconfiguration - Requires trained, qualified supervisor to direct climbing operations.
 BC CRANE SAFETY OPERATOR STANDARD	Section 6 – Top & Bottom Climbing <i>Operators must work under the direction of a qualified supervisor who understands the manufacturer's procedures.</i>
 SKILLEDTRADESBC PROGRAM	Tower Crane Operator Program Training aligned to Red Seal RSOS and WorkSafeBC regulatory requirements.



KEY POINT: A qualified supervisor has the training, knowledge and experience for the particular crane and activity and must be present and direct the climbing operation at all times.





TASK C-7: CLIMBING AND RECONFIGURATION

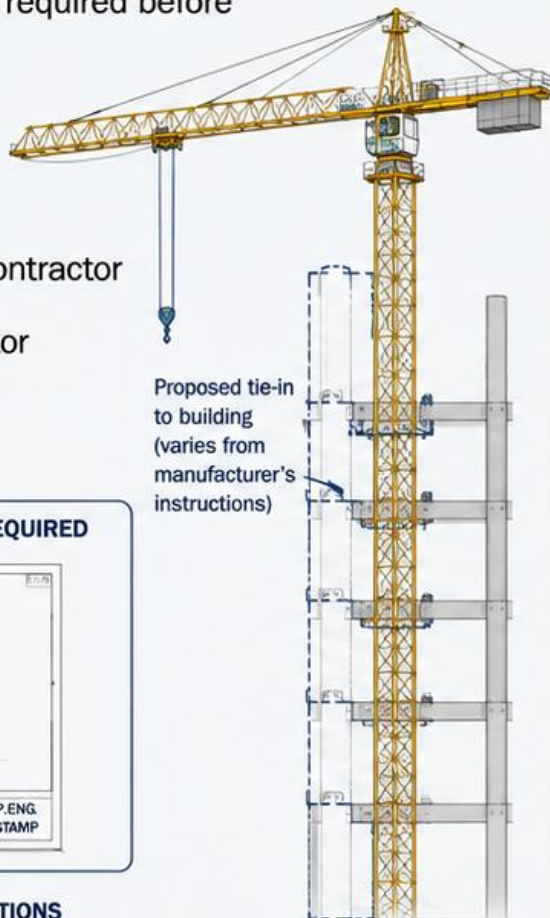
BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 3 – ANSWER

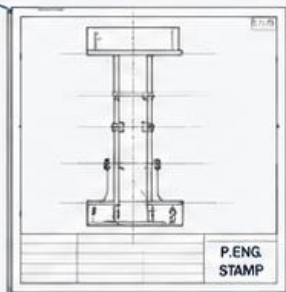
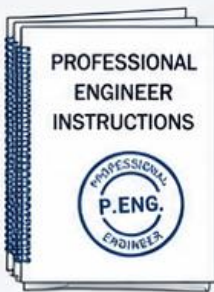
QUESTION 3

A proposed tower crane installation varies from the manufacturer's instructions. What is required before the climbing operation proceeds?

- ✓ **A. Written instructions from a professional engineer**
- B. Permission from the general contractor
- C. Approval from the crane operator
- D. A second qualified supervisor



ENGINEERED INSTALLATION DOCUMENTS REQUIRED



**DEVIATION FROM MANUFACTURER'S INSTRUCTIONS
REQUIRES PROFESSIONAL ENGINEER INSTRUCTIONS.**



CORRECT ANSWER: A

WRITTEN INSTRUCTIONS FROM A PROFESSIONAL ENGINEER

WorkSafeBC OHS Regulation **14.73.2(1)(b)** requires that if the installation varies from the manufacturer's instructions, written instructions from a professional engineer are required before erection, climbing, repositioning or dismantling proceeds.



**ENGINEER'S
INSTRUCTIONS
REQUIRED**

WHY THE OTHER CHOICES ARE WRONG

- B.** General contractor permission does not satisfy the regulatory requirement for professional engineer instructions.
- C.** operator approval is not a regulatory requirement and does not address engineering design responsibility.
- D.** A second qualified supervisor does not replace the need for professional engineer instructions when installation varies from the manufacturer's instructions.

REFERENCE BASIS

 WORKSAFEBC REGULATION	OHS Regulation, Part 14 – Cranes and Hoists 14.73.2(1)(b) <i>If the installation varies from the manufacturer's instructions, written instructions from a professional engineer are required.</i>
 RED SEAL RSOS (2023)	C-7.00 Climbing and Reconfiguration <i>Requires compliance with manufacturer or engineer instructions where applicable.</i>
 BC CRANE SAFETY OPERATOR STANDARD	Section 6 – Top & Bottom Climbing <i>Follow manufacturer's procedures. Deviations require professional engineer's instructions.</i>
 SKILLEDTRADESBC PROGRAM	Tower Crane Operator Program <i>Training aligned to Red Seal RSOS and WorkSafeBC requirements.</i>



KEY POINT: Any deviation from the manufacturer's installation instructions must be addressed by a professional engineer's written instructions before work begins.





TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 4 – ANSWER

QUESTION 4

A Notice of Project for tower crane climbing must normally be submitted to WorkSafeBC:

- A. 24 hours before the climb
- B. 72 hours before the climb
- C. One week before the climb
- ✓ D. At least two weeks before the climb



NOTICE OF PROJECT – CLIMBING



Submit at least two weeks (14 days) before the climbing operation begins.



CORRECT ANSWER: D AT LEAST TWO WEEKS BEFORE THE CLIMB

WorkSafeBC OHS Regulation **14.23(5)** requires that a Notice of Project must be submitted to WorkSafeBC at least two weeks before the commencement of erection, climbing, repositioning or dismantling of a tower crane.



NOTICE OF
PROJECT
REQUIRED

WHY THE OTHER CHOICES ARE WRONG

- A. Twenty-four hours is not a sufficient notice period under the regulation.
- B. Seventy-two hours (3 days) is not the required minimum notice period.
- C. One week (7 days) is less than the required two weeks (14 days).

REFERENCE BASIS



WORKSAFEBC
REGULATION

- OHS Regulation, Part 14 – Cranes and Hoists
14.23(5)
A notice of project must be submitted to WorkSafeBC at least two weeks before the commencement of erection, climbing, repositioning or dismantling of a tower crane.



RED SEAL RSOS
(2023)

- C-7.00 Climbing and Reconfiguration
Knowledge of applicable legislation and regulatory requirements.



BC CRANE SAFETY
OPERATOR STANDARD

- Section 1 – Legal & Regulatory Requirements
Operators must comply with applicable legislative requirements, including Notice of Project timelines.



SKILLEDTRADESBC
PROGRAM

- Tower Crane Operator Program
Training aligned to Red Seal RSOS and WorkSafeBC requirements.



KEY POINT: A Notice of Project must be submitted to WorkSafeBC at least two weeks (14 days) before any tower crane climbing operation begins.





TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 5 – ANSWER

QUESTION 5

The climbing crew arrives at the crane, but the applicable manufacturer's climbing instructions are not available at the workplace.

What should happen?

- A** Use the instructions for a similar crane.
- B** Continue if the supervisor has climbed this model previously.
- C** Obtain the applicable instructions before climbing proceeds. ✓
- D** Allow the most experienced crew member to determine the sequence.

APPLICABLE MANUFACTURER'S CLIMBING INSTRUCTIONS

Required to determine the correct climbing method, sequence, equipment, load limits, balance requirements and safety precautions.

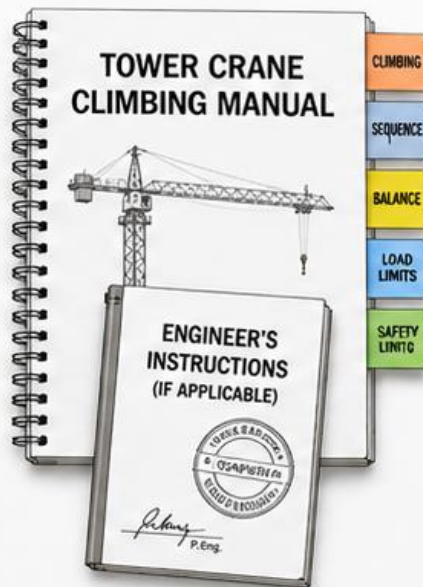


✗ DO NOT PROCEED WITHOUT THE APPLICABLE INSTRUCTIONS

Using the wrong sequence or method can lead to instability, component damage, or collapse.



OBTAIN INSTRUCTIONS



✓ REVIEW AND FOLLOW THE APPLICABLE INSTRUCTIONS

Ensure the correct method, sequence, load limits and safety precautions are followed.



CORRECT ANSWER: C
OBTAIN THE APPLICABLE INSTRUCTIONS BEFORE CLIMBING PROCEEDS.

Climbing must be performed in accordance with the applicable manufacturer's instructions (or professional engineer's instructions if those are required due to a deviation). These instructions provide the correct method, sequence, equipment, load limits, balance requirements and safety precautions.

Work should not begin until the required instructions are obtained, reviewed and understood by all personnel.

WHY THE OTHER CHOICES ARE WRONG

- A** Instructions for a similar crane may not be applicable to this model or configuration.
- B** Previous experience does not replace the requirement to follow the applicable instructions.
- D** The most experienced crew member cannot replace the required manufacturer's or engineer's instructions.

REFERENCE BASIS

WORK SAFE BC	WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists 14.73.2(1) <i>Climbing must be done in accordance with the manufacturer's instructions, or professional engineer's instructions where applicable.</i>
RSOS	Red Seal RSOS (2023) C-7.02 <i>Top Climbing – Raising and lowering procedures and instructions.</i>
BC	BC Crane Safety Operator Standard Section N2 – <i>Use and interpret OEM raising procedures, sequence, balance requirements, component inspections and wind limitations.</i>



KEY POINT: Do not proceed with a climb until the applicable manufacturer's climbing instructions (or engineer's instructions if required) are on site, reviewed and understood by all personnel. They control the method, sequence and safety requirements.





TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 6 – ANSWER

QUESTION 6

A tower crane has been climbed. Which **BEST** describes what is required before it is returned to service?

- A** Only the operator's normal pre-use inspection is required.
- B** The crane may operate below 50% capacity until documentation is completed.
- C** Required supervisor verification, applicable engineering certification, overload-prevention-system adjustment and required documentation must be completed.
- D** An unloaded hoist, trolley and slew test is sufficient.

REQUIREMENTS BEFORE RETURNING A CLIMBED CRANE TO SERVICE

After a climb, multiple requirements must be satisfied and documented before the crane can be returned to service.



CORRECT ANSWER: C

REQUIRED SUPERVISOR VERIFICATION, APPLICABLE ENGINEERING CERTIFICATION, OVERLOAD-PREVENTION-SYSTEM ADJUSTMENT AND REQUIRED DOCUMENTATION MUST BE COMPLETED.

WorkSafeBC OHSR 14.75 requires the qualified supervisor to verify compliance after climbing. If the installation varies from the manufacturer's instructions, the work must be carried out in accordance with instructions from a professional engineer (14.73.2(1)(b)). The overload-prevention system must be adjusted as necessary, and required records/certifications must be available before the crane is returned to service.

WHY THE OTHER CHOICES ARE WRONG

- A** A normal pre-use inspection alone does not satisfy 14.75 requirements.
- B** Operating below a percentage of capacity is not permitted until all requirements are met and verified.
- D** Function tests are only one part of the overall requirements.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.75 After climbing, the qualified supervisor must verify compliance with the applicable specifications before the crane is returned to service.



Red Seal RSOS (2023)
C-7.02 Top Climbing – Raising and lowering procedures and instructions.



BC Crane Safety Operator Standard
Section N2 – Post-climb inspections, setup of LMI, documentation and return-to-service requirements.



KEY POINT: A climbed crane is not ready for service until all required steps are complete: supervisor verification, engineering certification (if applicable), LMI adjustment, and documentation. These requirements ensure the crane is safe, stable and within rated capacity.





TASK C-7: CLIMBING AND RECONFIGURATION

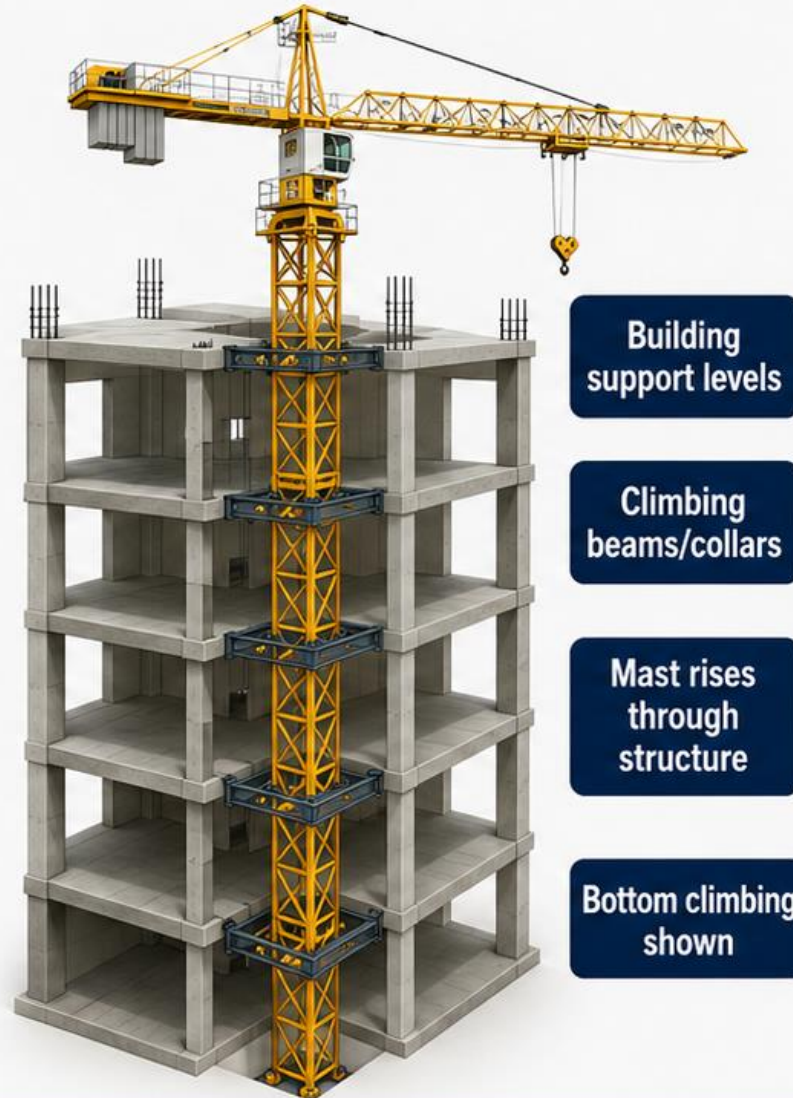
BC Red Seal Tower Crane Operator Practice Test

QUESTION 7 – ANSWER

QUESTION 7

Which statement **BEST** describes bottom climbing?

- A** New tower sections are inserted immediately below the upper works
- B** The crane and mast are raised and supported using beams/collars associated with the surrounding building structure 
- C** The jib is raised to increase hook height
- D** The crane climbs by increasing counterweight



CORRECT ANSWER: B — BOTTOM CLIMBING

Bottom climbing raises the crane and mast through the structure using engineered building supports such as beams, collars or support frames.

It is different from top climbing, where new mast sections are inserted below the upper works.

WHY THE OTHER CHOICES ARE WRONG

- A** A = describes top climbing, not bottom climbing.
- C** C = raising the jib does not climb the tower.
- D** D = counterweight does not make the crane climb.

REFERENCE BASIS



Red Seal RSOS 2023 — C-7.01 Bottom Climbing



WorkSafeBC Part 14 — engineered supports / crane support requirements



BC Crane Safety — bottom-climbing competency



KEY POINT: Bottom climbing depends on engineered building supports. Top climbing adds mast sections near the upper works.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 8 – ANSWER

QUESTION 8

Before a bottom-climbing operation, which condition is particularly important to verify?

- A** Building support locations and structural requirements ✓
- B** Hook-block paint condition
- C** Operator seat adjustment
- D** Trolley travel speed

BOTTOM-CLIMBING PREPARATION – WHAT TO VERIFY

Bottom climbing relies on the building structure to support the crane as it is raised. Verify the critical items before starting the climb.

A. BUILDING SUPPORT LOCATIONS & STRUCTURAL REQUIREMENTS



Verify support locations, climbing beams, collars, pins, shims and connections are installed as per the engineering drawings and that the structure can support the required loads.

✓ **CRITICAL – VERIFY FIRST**

B. HOOK-BLOCK PAINT CONDITION



Check hook block for paint condition, cracks, wear and identification. Important, but not the primary focus before bottom climbing.

i **IMPORTANT, NOT PRIMARY**

C. OPERATOR SEAT ADJUSTMENT



Adjust seat and controls for operator comfort and visibility. Important for operation, but not the primary focus before bottom climbing.

i **IMPORTANT, NOT PRIMARY**

D. TROLLEY TRAVEL SPEED



Verify trolley travel speed as part of normal checks. Not the primary condition that ensures structural support during bottom climbing.

i **IMPORTANT, NOT PRIMARY**



CORRECT ANSWER: A
BUILDING SUPPORT LOCATIONS AND STRUCTURAL REQUIREMENTS.

Bottom climbing depends on the building structure to carry the crane loads during the climb. Verifying that support locations, climbing beams/collars and structural requirements are installed and capable of supporting the loads is the most critical condition to confirm before starting the operation.

WHY THE OTHER CHOICES ARE WRONG

- B** Paint condition is part of inspections but does not ensure structural support during climbing.
- C** Seat adjustment affects operator comfort and control but not structural safety during climbing.
- D** Trolley travel speed is operational and is not the primary safety condition for bottom climbing.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.74 *Bottom climbing and support systems must be installed in accordance with the manufacturer's and engineer's instructions.*



Red Seal RSOS (2023)
C-7.01 *Bottom Climbing – Verify building support arrangements and structural requirements before climbing.*



BC Crane Safety Operator Standard
Section N2 – Top & Bottom Climbing
Operators must verify that support systems and structure are as required before climbing operations begin.



KEY POINT: Bottom climbing is only as safe as the structure that supports the crane. Verify the building support locations and structural requirements first to ensure the crane is supported correctly throughout the climb.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 9 – ANSWER

QUESTION 9

Engineering documents require a specific climbing-beam/support arrangement. The actual installation does not match the engineering drawings. What should happen?

- A** Continue if the crane is unloaded
- B** Modify the arrangement using available material
- C** Stop and have the discrepancy resolved in accordance with the engineering requirements ✓
- D** Reduce climbing speed

DISCREPANCY BETWEEN ENGINEERING DOCUMENTS AND ACTUAL INSTALLATION

The installed climbing-beam/support arrangement must match the engineering design. Any discrepancy must be corrected before the climb proceeds.

A. CONTINUE IF THE CRANE IS UNLOADED



Unloaded does not eliminate the risk of structural failure or instability. Discrepancy remains unresolved.

B. MODIFY THE ARRANGEMENT USING AVAILABLE MATERIAL



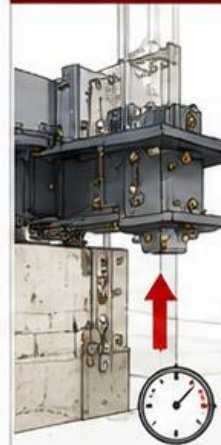
Unauthorized modifications can compromise the engineered design and may not have adequate strength or capacity.

C. STOP AND RESOLVE THE DISCREPANCY PER ENGINEERING REQUIREMENTS



Correct action. Stop the operation and have the discrepancy resolved in accordance with the engineering requirements before proceeding.

D. REDUCE CLIMBING SPEED



Lowering speed does not address the discrepancy. The risk of structural failure remains.



CORRECT ANSWER: C
STOP AND HAVE THE DISCREPANCY RESOLVED IN ACCORDANCE WITH THE ENGINEERING REQUIREMENTS.

All climbing supports, beams, collars and connections must be installed as shown and designed. Deviations create unknown stresses and can lead to loss of support, collapse, or instability. The discrepancy must be corrected and verified by a qualified person in accordance with the engineering requirements before the climb can continue.

WHY THE OTHER CHOICES ARE WRONG

- A** An unloaded crane can still create excessive forces or instability if the support arrangement is incorrect.
- B** Improvised modifications are not engineered and can introduce new hazards.
- D** Reducing speed does not correct an incorrect or inadequate support arrangement.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.74 Bottom climbing and support systems must be installed in accordance with the manufacturer's and engineer's instructions.



Red Seal RSOS (2023)
C-7.01 Bottom Climbing – Verify building support arrangements and structural requirements before climbing.



BC Crane Safety Operator Standard
Section N2 – Top & Bottom Climbing
Operators must verify that support systems and structure are as required before climbing operations begin.



KEY POINT: Structural supports must match the engineering design exactly. Do not improvise or accept deviations. Stop the operation and correct the discrepancy in accordance with the engineering requirements.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 10 – ANSWER

QUESTION 10

During a bottom climb, unexpected movement occurs at a structural support point while load is being transferred.

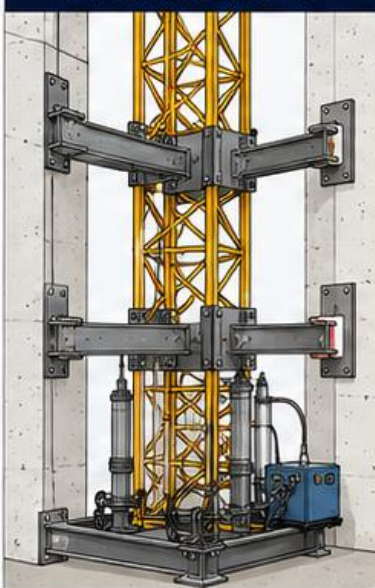
What is the BEST action?

- A** Increase hydraulic pressure to complete the transfer
- B** Stop the operation, establish a safe condition and investigate the movement 
- C** Continue until the next support level is reached
- D** Add temporary material beneath the support while continuing the climb

BOTTOM (INTERNAL) CLIMB – UNEXPECTED MOVEMENT DURING LOAD TRANSFER

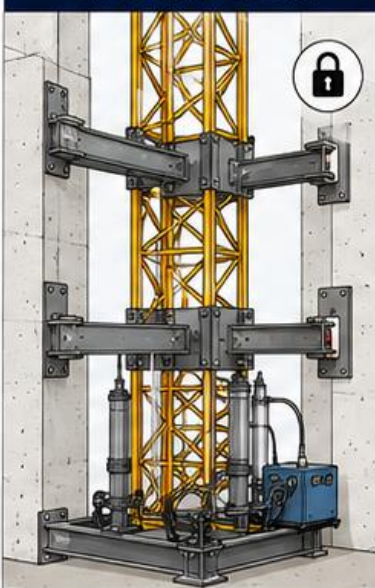
In a bottom climb, the crane and climbing assembly are raised using hydraulic equipment. Climbing collars are supported by steel beams bearing into the building structure. Any unexpected movement at a support point must be addressed immediately.

1. MOVEMENT DETECTED



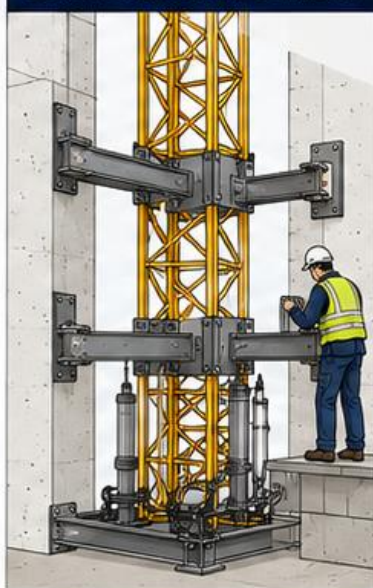
Unexpected movement or gap appears at a structural support point while load is being transferred.

2. STOP AND SECURE



Stop the climb immediately. Hold the load safely and establish a stable, secure condition. Keep all personnel clear of the hazard area.

3. INVESTIGATE AND CORRECT



Investigate the cause of the movement. Correct the issue in accordance with engineering requirements before any further climbing occurs.



CORRECT ANSWER: B

STOP THE OPERATION, ESTABLISH A SAFE CONDITION AND INVESTIGATE THE MOVEMENT.

Unexpected movement during a load transfer indicates a potential structural issue or instability. Continuing the climb can lead to loss of support, collapse or serious injury. The operation must be stopped, the load secured and the cause of the movement investigated and corrected before any further climbing occurs.

WHY THE OTHER CHOICES ARE WRONG

- A** Increasing hydraulic pressure can worsen the movement or cause structural failure.
- C** Continuing the climb ignores the hazard and increases the risk of collapse or loss of support.
- D** Adding temporary material without engineering direction or investigation may not address the root cause and can create additional hazards.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.74 Bottom climbing and support systems must be installed in accordance with the manufacturer's and engineer's instructions.



Red Seal RSOS (2023)
C-7.01 Bottom Climbing – Verify building support arrangements and structural requirements before climbing.



BC Crane Safety Operator Standard
Section N2 – Top & Bottom Climbing
Operators must verify that support systems and structure are as required before climbing operations begin.



KEY POINT: Any unexpected movement during a climb is a warning sign. Stop immediately, secure the load, investigate the cause and correct the issue in accordance with the engineering requirements before continuing.





TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 11 – ANSWER

QUESTION 11

Which group contains components associated with top-climbing systems?

A Climbing unit, secured dogs, hydraulic components, collars and beams



B Outrigger floats, axle locks and boom dolly

C Travel brakes, steering cylinders and tires

D Crawler tracks, boom stops and gantry

TOP-CLIMBING SYSTEM COMPONENTS

Top-climbing systems use a climbing unit that lifts the crane and is secured to the tower with dogs. The system is supported by collars and beams and powered by hydraulic components.

A. TOP-CLIMBING SYSTEM COMPONENTS



Climbing unit



Secured dogs



Hydraulic components



Collars



Beams

These components are used in top-climbing systems to raise and secure the crane.



B. MOBILE CRANE ACCESSORIES



Outrigger floats



Axle locks



Boom dolly

Used with mobile cranes for setup, transport and stability.



C. MOBILE CRANE DRIVE COMPONENTS



Travel brakes



Steering cylinders



Tires

These components are part of the crane's travel and steering systems.



D. CRAWLER CRANE COMPONENTS



Crawler tracks



Boom stops



Gantry

These components are associated with crawler cranes and lifting aids.



CORRECT ANSWER: A
CLIMBING UNIT, SECURED DOGS,
HYDRAULIC COMPONENTS, COLLARS
AND BEAMS

Top-climbing systems use a climbing unit with secured dogs that anchor to the tower. Hydraulic components power the climbing unit, and collars and beams transfer loads between the crane and the tower structure.

WHY THE OTHER CHOICES ARE WRONG

B Outrigger floats, axle locks and boom dolly are mobile crane accessories.

C Travel brakes, steering cylinders and tires are drive and steering components.

D Crawler tracks, boom stops and gantry are crawler crane components.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.74 Bottom climbing and support systems must be installed in accordance with the manufacturer's and engineer's instructions.



Red Seal RSOS (2023)
C-7.01 Bottom Climbing – Knowledge of climbing systems and associated components.



BCCSA COR® Audit – Cranes
Verify that climbing systems are installed, used and maintained in accordance with OEM instructions and engineering.



KEY POINT: Top-climbing systems include the climbing unit, secured dogs, hydraulic components, collars and beams. Know the components and their function.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 12 – ANSWER

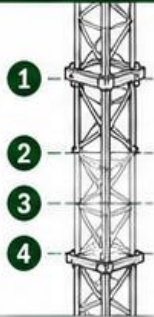
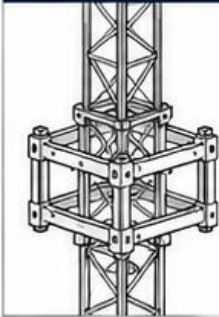
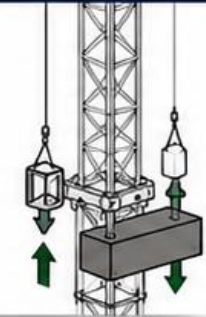
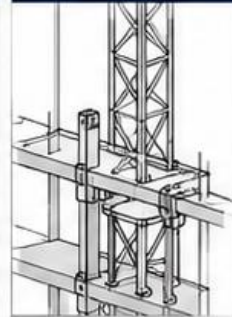
QUESTION 12

Which of the following is specifically identified in the Red Seal Occupational Standard as required top-climbing knowledge?

- A** Importance of the climbing sequence 
- B** Concrete pump output
- C** Mobile-crane outrigger configuration
- D** Truck axle loading

RED SEAL OCCUPATIONAL STANDARD – TOP-CLIMBING KNOWLEDGE

The Red Seal Occupational Standard for Tower Crane Operator (Top-slewing) identifies the following required knowledge related to top-climbing systems.

1. CLIMBING SEQUENCE	2. CLIMBING SYSTEM COMPONENTS	3. LOAD TRANSFER PRINCIPLES	4. SUPPORT AND ENGINEERING	5. SAFETY AND HAZARD CONTROL
				
Importance of the climbing sequence Explain and apply the correct order of operations for top-climbing to maintain stability and safety.	Identify and describe climbing units, collars, secured dogs, hydraulic components, and beams and their function.	Explain how loads are transferred between the crane and the structure during climbing operations.	Explain support requirements, engineering documentation and verification for top-climbing systems.	Identify hazards associated with top-climbing and describe the controls to perform the work safely.

-  **The STANDARD explicitly requires knowledge of the importance of the climbing sequence as part of the top-climbing knowledge requirements.**



CORRECT ANSWER: A IMPORTANCE OF THE CLIMBING SEQUENCE

The Red Seal Occupational Standard specifically identifies the importance of the climbing sequence as required top-climbing knowledge.

It is not related to concrete pumps, mobile-crane outrigger configurations or truck axle loading.

WHY THE OTHER CHOICES ARE WRONG

- B** Concrete pump output is not part of the Red Seal top-climbing knowledge requirements.
- C** Mobile-crane outrigger configuration is not part of the Red Seal top-climbing knowledge requirements.
- D** Truck axle loading is not part of the Red Seal top-climbing knowledge requirements.

REFERENCE BASIS



Red Seal RSOS (2023)
C-7.02 Top Climbing – Required knowledge includes, but is not limited to: climbing sequence, climbing system components, load transfer principles, support and engineering, and safety and hazard control.



BCCSA COR® Audit – Cranes
Verify that top-climbing operations are performed in accordance with manufacturer's instructions, engineering documentation and safe work practices.



KEY POINT: The Red Seal OS requires knowledge of the importance of the climbing sequence for top-climbing. Know the correct order of operations and why it is critical to safe climbing.





TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 13 – ANSWER

QUESTION 13

Before top climbing begins, required shoring has not been completed.

What should happen?

- A** Begin the climb and complete the shoring later
- B** Proceed because the hook is unloaded
- C** Do not begin until the required shoring is in place and meets the applicable requirements 
- D** Reduce hydraulic pressure during the climb

TOP CLIMBING – REQUIRED SHORING

Top climbing places loads into the building structure. Required shoring must be installed and meet the engineering requirements before the climb begins.
Never assume the structure will carry the loads without proper shoring.

NOT ACCEPTABLE SHORING NOT IN PLACE



Do not begin climbing.
Required shoring is missing.
Loads could cause structural failure, collapse or serious injury.

NOT ACCEPTABLE INCOMPLETE / NON-COMPLIANT SHORING



Do not begin climbing.
Shoring is incomplete or does not meet the engineering requirements.
Correct before proceeding.

ACCEPTABLE SHORING IN PLACE AND COMPLIANT



Begin climbing only when all required shoring is installed and meets the engineering requirements.
Verify before the climb.



CORRECT ANSWER: C

DO NOT BEGIN UNTIL THE REQUIRED SHORING IS IN PLACE AND MEETS THE APPLICABLE REQUIREMENTS

Top-climbing loads must be safely transferred into the structure through properly installed shoring.

Climbing must not begin until the shoring is complete and compliant with the engineering design and applicable codes.

WHY THE OTHER CHOICES ARE WRONG

- A** Beginning the climb before shoring is in place can overstress the structure and cause collapse.
- B** The load on the hook is not the only load being transferred; the crane and climber impose significant loads on the structure.
- D** Reducing hydraulic pressure does not address the lack of structural support and does not make the operation safe.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.74 Bottom climbing and support systems must be installed in accordance with the manufacturer's and engineer's instructions.



Red Seal RSOS (2023)
C-7.02 Top Climbing – Required knowledge includes climbing system components, support and engineering, and safety.



BC Crane Safety Operator Standard
Section N2 – Top & Bottom Climbing
Operators must verify that support systems and structure are as required before climbing operations begin.



KEY POINT: Never start a top-climb until all required shoring is installed, inspected and compliant.
Verify against the engineering drawings and manufacturer's instructions.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 14 – ANSWER

QUESTION 14

Which group contains weather conditions specifically identified in the RSOS as affecting climbing?

A Wind, freezing rain, temperature, snow, lightning and fog



B Wind and temperature only

C Rain, humidity and barometric pressure

D Wind, UV index and humidity

WEATHER CONDITIONS SPECIFICALLY IDENTIFIED IN THE RSOS AS AFFECTING CLIMBING

The Red Seal Occupational Standard (RSOS) for Tower Crane Operator (Top-slewing) identifies the following weather conditions as affecting climbing operations.

WIND	FREEZING RAIN	TEMPERATURE	SNOW	LIGHTNING	FOG
					
High wind can cause increased loads and sway, affecting stability and control during climbing.	Freezing rain creates ice accumulation on structures and components, reducing grip and increasing the risk of slipping or sticking.	Extreme cold or heat can affect equipment performance, hydraulic systems, materials and operator judgement.	Snow accumulation adds weight, reduces visibility and can affect footing and access for climbing operations.	Lightning presents a significant risk to personnel and equipment; climbing must stop when lightning is present or in the area.	Fog reduces visibility, making it difficult to assess clearances, alignment and positions during climbing.



The RSOS specifically identifies wind, freezing rain, temperature, snow, lightning and fog as weather conditions that affect climbing operations.



CORRECT ANSWER: A
WIND, FREEZING RAIN, TEMPERATURE, SNOW, LIGHTNING AND FOG

The Red Seal Occupational Standard specifically identifies wind, freezing rain, temperature, snow, lightning and fog as weather conditions that affect climbing operations.

These conditions must be monitored and assessed before and during climbing.

WHY THE OTHER CHOICES ARE WRONG

- B** The RSOS includes more than just wind and temperature.
- C** Rain, humidity and barometric pressure are not the specific set identified in the RSOS for climbing.
- D** UV index and humidity are not part of the RSOS specific list for climbing.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.74 Bottom climbing and support systems must be installed in accordance with the manufacturer's and engineer's instructions.



Red Seal RSOS (2023)
C-7.02 Top Climbing – Required knowledge includes environmental conditions that affect climbing safety.



BCCSA COR® Audit – Cranes
Verify that weather conditions affecting climbing are assessed and monitored in accordance with procedures.



KEY POINT: The RSOS specifically identifies wind, freezing rain, temperature, snow, lightning and fog as weather conditions that affect top climbing. Assess and monitor these conditions before and during climbing.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 15 – ANSWER

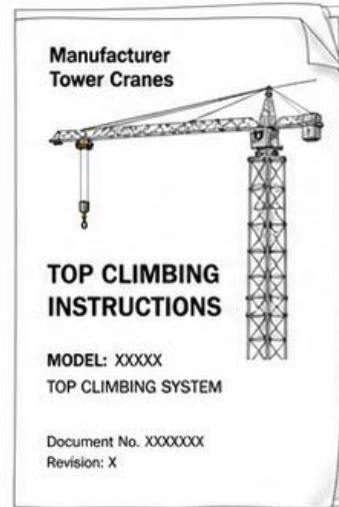
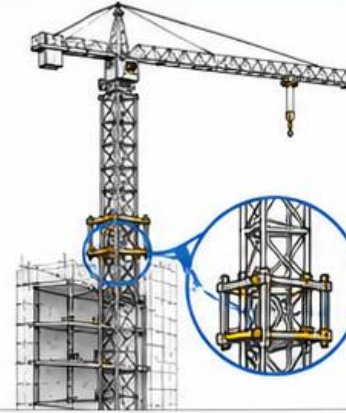
QUESTION 15

Before top climbing, where should the crew determine the applicable maximum climbing wind speed?

- A** A universal WorkSafeBC climbing limit of 37 km/h
- B** The previous day's recorded wind speed
- C** The applicable manufacturer's climbing instructions/specifications 
- D** WorkSafeBC's general 50 km/h fallback operating limit

DETERMINE THE APPLICABLE MAXIMUM CLIMBING WIND SPEED

Maximum climbing wind speed is **NOT** a universal value. It must be obtained from the applicable manufacturer's climbing instructions/specifications for the specific crane model, configuration and climbing system.



6.0 CLIMBING OPERATIONS

6.4 WEATHER CONDITIONS

Do not climb if wind speed exceeds the maximum allowable climbing wind speed specified for this crane and configuration.

MAXIMUM ALLOWABLE CLIMBING WIND SPEED (3-SECOND GUST)

Refer to model and configuration tables below.

Crane Model / Configuration	Max. Climbing Wind Speed
XXXXX (Top Climbing System A)	Refer to table
XXXXX (Top Climbing System B)	Refer to table
...	...

Values are in km/h (3-second gust).



Exceeding the specified climbing wind speed can result in loss of stability, structural damage, or serious injury.



CORRECT ANSWER: C
THE APPLICABLE MANUFACTURER'S CLIMBING INSTRUCTIONS/SPECIFICATIONS

The maximum climbing wind speed varies by crane model, configuration and climbing system.

It must be obtained from the applicable manufacturer's climbing instructions/specifications before climbing begins.

WHY THE OTHER CHOICES ARE WRONG

- A** There is no universal WorkSafeBC climbing wind limit such as 37 km/h.
- B** Previous recorded wind speeds do not determine the maximum allowable climbing wind speed.
- D** The general 50 km/h operating limit is not the applicable maximum climbing wind speed and may be lower or higher depending on the crane and system.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.74 Bottom climbing and support systems must be installed in accordance with the manufacturer's and engineer's instructions.



Red Seal RSOS (2023)
C-7.02 Top Climbing – Required knowledge includes weather conditions affecting climbing operations.



BCCSA COR® Audit – Cranes
Verify that weather conditions affecting climbing are assessed and monitored in accordance with manufacturer's instructions and safe work practices.



KEY POINT: Always obtain the maximum climbing wind speed from the applicable manufacturer's climbing instructions/specifications for the specific crane model, configuration and climbing system.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 15 – ANSWER

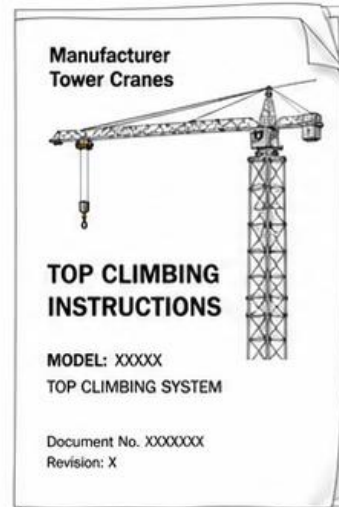
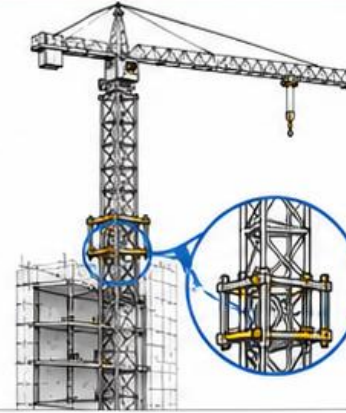
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DETERMINE THE APPLICABLE MAXIMUM CLIMBING WIND SPEED

Maximum climbing wind speed is **NOT** a universal value. It must be obtained from the applicable manufacturer's climbing instructions/specifications for the specific crane model, configuration and climbing system.



6.0 CLIMBING OPERATIONS

6.4 WEATHER CONDITIONS

Do not climb if wind speed exceeds the maximum allowable climbing wind speed specified for this crane and configuration.

MAXIMUM ALLOWABLE CLIMBING WIND SPEED (3-SECOND GUST)

Refer to model and configuration tables below.

Crane Model / Configuration	Max. Climbing Wind Speed
XXXXX (Top Climbing System A)	Refer to table
XXXXX (Top Climbing System B)	Refer to table
...	...

Values are in km/h (3-second gust).



Exceeding the specified climbing wind speed can result in loss of stability, structural damage, or serious injury.



CORRECT ANSWER: C
THE APPLICABLE MANUFACTURER'S
CLIMBING INSTRUCTIONS/SPECIFICATIONS

The maximum climbing wind speed varies by crane model, configuration and climbing system.

It must be obtained from the applicable manufacturer's climbing instructions/specifications before climbing begins.

WHY THE OTHER CHOICES ARE WRONG

- A** There is no universal WorkSafeBC climbing wind limit such as 37 km/h.
- B** Previous recorded wind speeds do not determine the maximum allowable climbing wind speed.
- D** The general 50 km/h operating limit is not the applicable maximum climbing wind speed and may be lower or higher depending on the crane and system.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.74 Bottom climbing and support systems must be installed in accordance with the manufacturer's and engineer's instructions.



Red Seal RSOS (2023)
C-7.02 Top Climbing – Required knowledge includes weather conditions affecting climbing operations.



BCCSA COR® Audit – Cranes
Verify that weather conditions affecting climbing are assessed and monitored in accordance with manufacturer's instructions and safe work practices.



 **KEY POINT:** Always obtain the maximum climbing wind speed from the applicable manufacturer's climbing instructions/specifications for the specific crane model, configuration and climbing system.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 16 – ANSWER

QUESTION 16

The ambient temperature drops below -18°C (0°F) during tower crane operations in British Columbia.

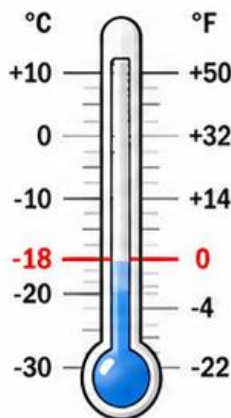
What applies?

- A** Operations may continue at half capacity
- B** Operations must stop unless otherwise specified by the manufacturer or a professional engineer
- C** Only climbing must stop
- D** Operations may continue if hydraulic-oil temperature remains above 0°C

COLD WEATHER REQUIREMENT – BRITISH COLUMBIA

In British Columbia, when the ambient temperature is below -18°C (0°F), tower crane **operations must stop unless otherwise specified by the manufacturer or a professional engineer.**

AMBIENT TEMPERATURE



-18°C (0°F)
Threshold

WHAT THIS MEANS



All tower crane operations must stop.

Do not operate unless otherwise specified by the manufacturer or a professional engineer.

"OTHERWISE SPECIFIED" BY:

1. MANUFACTURER



Manufacturer's written instructions or specifications for the specific crane model and configuration.

2. PROFESSIONAL ENGINEER



Written direction or certification from a Professional Engineer.



This requirement is intended to prevent brittle fracture of structural steel, hydraulic component failure, and loss of control in extreme cold.



CORRECT ANSWER: B

OPERATIONS MUST STOP UNLESS OTHERWISE SPECIFIED BY THE MANUFACTURER OR A PROFESSIONAL ENGINEER

When ambient temperature drops below -18°C (0°F) in British Columbia, all tower crane operations must stop unless otherwise specified by the manufacturer or a professional engineer.

WHY THE OTHER CHOICES ARE WRONG

- A** There is no provision to allow operations to continue at half capacity below -18°C (0°F). Operations must stop.
- C** The requirement applies to all tower crane operations, not just climbing.
- D** Hydraulic-oil temperature is not the controlling factor. The rule is based on ambient temperature.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80 Ambient temperature below -18°C (0°F)

(1) A tower crane must not be operated when the ambient temperature is below -18°C (0°F) unless the manufacturer or a professional engineer otherwise specifies.



Red Seal RSOS (2023)
C-7.01 Top Climbing – Required knowledge includes environmental conditions that affect climbing and crane operations.



BCCSA COR® Audit – Cranes
Verify that environmental conditions are assessed and that operations cease or are modified in accordance with regulatory and manufacturer requirements.



KEY POINT: When ambient temperature in B.C. drops below -18°C (0°F), stop all tower crane operations unless the manufacturer or a professional engineer has otherwise specified.





TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 17 – ANSWER

QUESTION 17

Thunder is heard while a climbing operation is underway.

What is the BEST response?

- A** Continue until lightning becomes visible
- B** Complete the mast section before stopping
- C** Stop and secure the operation as required and have personnel seek appropriate shelter 
- D** Continue because the tower crane is grounded

THUNDER HEARD DURING CLIMBING OPERATIONS

When thunder is heard, the storm is close enough for lightning to strike.



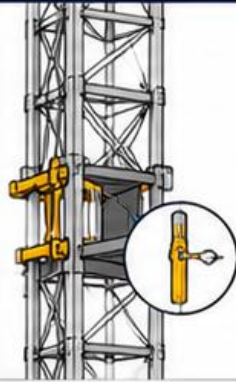
Stop the climbing operation immediately and protect personnel.

1. STOP THE CLIMB



Cease all climbing movements immediately. Do not continue to climb or rush to complete any section.

2. SECURE THE OPERATION



Secure the crane and climbing equipment as required by the manufacturer. Ensure loads are safe and stable.

3. MOVE PERSONNEL TO SHELTER



Direct all personnel to leave the crane and seek appropriate shelter immediately. Do not remain on or near the structure.

4. WAIT UNTIL IT IS SAFE



Remain in shelter until at least 30 minutes after the last clap of thunder before resuming operations.



**Thunder can occur without visible lightning.
When thunder is heard, act immediately.**



CORRECT ANSWER: C

**STOP AND SECURE THE OPERATION AS REQUIRED
AND HAVE PERSONNEL SEEK APPROPRIATE SHELTER**

When thunder is heard, lightning is close enough to be a serious threat.

The best response is to stop and secure the operation as required and have personnel seek appropriate shelter.

This protects personnel from lightning strike and prevents injury or equipment damage.

WHY THE OTHER CHOICES ARE WRONG

- A** Lightning can strike before it is seen.
- B** Completing the mast section increases exposure time and risk.
- D** While the crane is grounded, lightning can still strike the structure or nearby and travel through it.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80(1)(b) When thunder is heard or lightning observed in the area, crane operations must be stopped.



Red Seal RSOS (2023)
C-7.01 Top Climbing – Required knowledge includes environmental conditions that affect climbing and crane operations.



BCCSA COR® Audit – Cranes
Verify that operations are suspended during electrical storms and that personnel follow safe work practices.



KEY POINT: When thunder is heard, lightning is close enough to strike.

Stop, secure, and get to shelter. Wait 30 minutes after the last thunder before resuming work.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 18 – ANSWER

QUESTION 18

Why is crane balance critical during top climbing?

A It increases the crane's rated hook capacity

B It controls forces acting on the climbing system and helps maintain stability



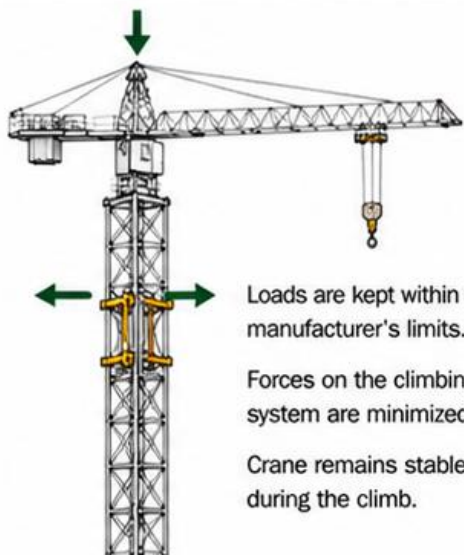
C It increases hydraulic-system speed

D It prevents normal trolley movement

CRANE BALANCE DURING TOP CLIMBING

Maintaining proper crane balance is essential to control the forces on the climbing system and maintain overall stability of the crane and structure.

BALANCED – STABLE

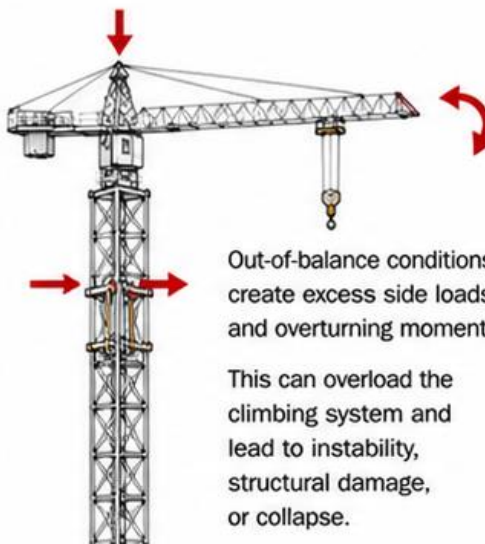


Loads are kept within the manufacturer's limits.

Forces on the climbing system are minimized.

Crane remains stable during the climb.

UNBALANCED – UNSTABLE



Out-of-balance conditions create excess side loads and overturning moments.

This can overload the climbing system and lead to instability, structural damage, or collapse.



Proper balance keeps forces within design limits, protects the climbing system, and ensures the stable and safe completion of each climb.



CORRECT ANSWER: B

IT CONTROLS FORCES ACTING ON THE CLIMBING SYSTEM AND HELPS MAINTAIN STABILITY

Crane balance during top climbing controls the forces on the climbing system and helps maintain stability of the crane and structure.

Out-of-balance conditions increase loads on the climbing system and can lead to instability, damage, or collapse.

WHY THE OTHER CHOICES ARE WRONG

A Balance does not increase the rated hook capacity.

C Balance has no effect on hydraulic-system speed.

D Balance does not prevent normal trolley movement; it ensures safe climbing within design limits.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.02 Top Climbing – Required knowledge includes balance, stability, and control of forces during climbing.



BCCSA COR® Audit – Cranes
Verify that top climbing is planned and performed in accordance with manufacturer's instructions to maintain stability.



KEY POINT: Proper balance is critical during top climbing because it controls forces on the climbing system and helps maintain stability of the crane, structure, and personnel.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 19 – ANSWER

QUESTION 19

The manufacturer specifies a particular trolley position or luffing configuration to establish climbing balance. What should the operator do?

- A** Establish the manufacturer's specified configuration 
- B** Position the hook at minimum radius regardless of the procedure
- C** Use the configuration from the previous crane climbed
- D** Select whichever position results in the lowest hydraulic pressure

FOLLOW THE MANUFACTURER'S SPECIFIED CONFIGURATION

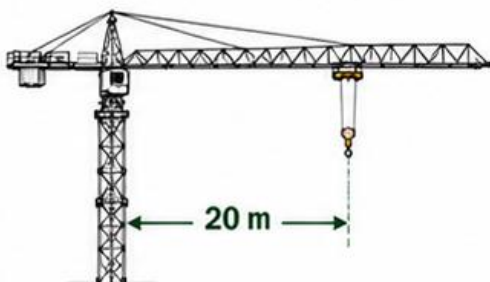
The manufacturer provides a specific trolley position or luffing configuration to establish the correct climbing balance.

The operator must establish the manufacturer's specified configuration before initiating the climb.

FOLLOW THE SPECIFIED CONFIGURATION

Example – Flat-Top Crane

Manufacturer specifies trolley position at 20 m radius.



 Proper balance is achieved. Climb can proceed safely.

 Always follow the manufacturer's instructions for the specific crane model, climbing system, and configuration.

DO NOT GUESS OR DEVIATE

 Minimum radius not always correct. Can create an unbalanced condition.



 Previous crane may be different. Each crane model and configuration is unique.



 Lowest hydraulic pressure is not the goal. Balance and stability are the priority.



CORRECT ANSWER: A ESTABLISH THE MANUFACTURER'S SPECIFIED CONFIGURATION

The manufacturer specifies the exact trolley position or luffing configuration required to achieve proper climbing balance for that crane and climbing system. The operator must establish the manufacturer's specified configuration before beginning the climb.

WHY THE OTHER CHOICES ARE WRONG

- B** Minimum radius is not always the correct position and can create an unbalanced condition.
- C** Each crane and climbing system is different. The previous crane's configuration may not apply.
- D** Lowest hydraulic pressure is not the objective. Balance and stability are the priority.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.03 Top Climbing – Required knowledge includes following manufacturer's instructions and ensuring proper balance prior to climbing.



BCCSA COR® Audit – Cranes
Verify that top climbing operations are performed in accordance with manufacturer's instructions to maintain stability.



KEY POINT: Establish the manufacturer's specified trolley position or luffing configuration to achieve proper climbing balance before initiating the climb.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 20 – ANSWER

QUESTION 20

While establishing the climbing condition, the crane cannot be balanced as required by the manufacturer's procedure. What should happen?

- A** Increase hydraulic pressure
- B** Add temporary counterweight
- C** Stop and resolve the balance condition in accordance with the applicable procedure 
- D** Continue slowly until the next support point is reached

CRANE CANNOT BE BALANCED – STOP AND RESOLVE

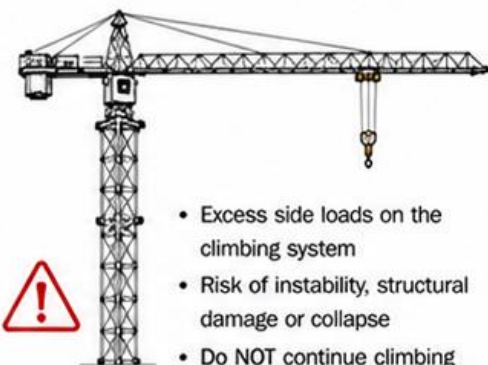
If the crane cannot be balanced as required by the manufacturer's procedure, do **NOT** proceed.

Stop the climbing operation and resolve the balance condition in accordance with the applicable procedure.



UNBALANCED CONDITION – NOT PERMITTED

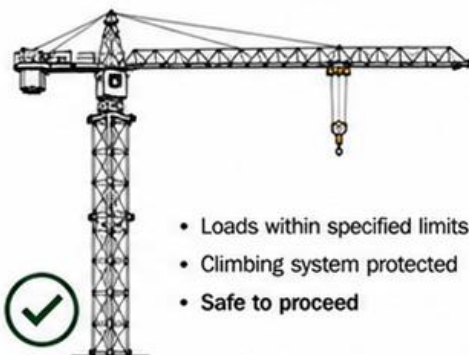
Forces are outside the limits specified by the manufacturer's procedure.



- Excess side loads on the climbing system
- Risk of instability, structural damage or collapse
- Do NOT continue climbing

BALANCED CONDITION – PERMITTED

Forces are within the limits specified by the manufacturer's procedure.



- Loads within specified limits
- Climbing system protected
- Safe to proceed



Balance is not achieved by forcing the system. Stop, assess, and take the required corrective actions in accordance with the manufacturer's instructions.



CORRECT ANSWER: C

STOP AND RESOLVE THE BALANCE CONDITION IN ACCORDANCE WITH THE APPLICABLE PROCEDURE

If the crane cannot be balanced as required by the manufacturer's procedure, the operation must stop.

The balance condition must be resolved in accordance with the applicable procedure before climbing can safely continue.

WHY THE OTHER CHOICES ARE WRONG

- A** Increasing hydraulic pressure can increase loads and risk, and will not correct an unbalanced condition.
- B** Adding temporary counterweight is not permitted unless specifically allowed by the manufacturer's procedure.
- D** Continuing to climb an unbalanced crane can lead to instability, structural damage or collapse.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists 14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023) C-7.03 Top Climbing – Required knowledge includes balance, stability, and control of forces during climbing.



BCCSA COR® Audit – Cranes Verify that top climbing is planned and performed in accordance with manufacturer's instructions to maintain stability.



KEY POINT: If the crane cannot be balanced as required, stop the climb and resolve the condition in accordance with the applicable procedure. Do not continue until balance is achieved.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 21 – ANSWER

QUESTION 21

Before additional components are placed into service during a tower crane reconfiguration, what inspection does the RSOS specifically require the operator to verify has been completed?

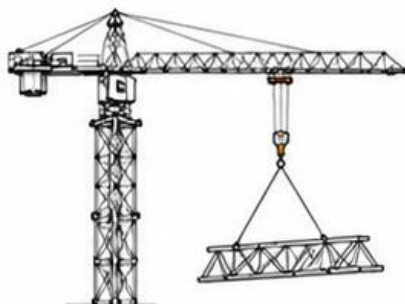
- A** Magnetic inspection of every tower bolt
- B** Required non-destructive testing (NDT) of the additional components 
- C** Annual inspection of the entire crane
- D** Daily inspection by the prime contractor

RSOS REQUIREMENT – NDT OF ADDITIONAL COMPONENTS

Before additional components are placed into service during a tower crane reconfiguration, the RSOS requires the operator to verify that the required non-destructive testing (NDT) of those additional components has been completed.



RSOS C-7.04 RECONFIGURATION – ADDITIONAL COMPONENTS



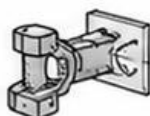
VERIFICATION REQUIRED BY OPERATOR

The operator must verify that the required **non-destructive testing (NDT)** of the additional components has been completed before those components are placed into service.

COMMON COMPONENTS THAT MAY REQUIRE NDT



Mast sections



Jib sections



Slewing platform components



Structural pins and connections



Other structural components as specified



NDT is used to detect cracks, internal defects, or other damage that may not be visible but could compromise the component's integrity.



CORRECT ANSWER: B

REQUIRED NON-DESTRUCTIVE TESTING (NDT) OF THE ADDITIONAL COMPONENTS

The RSOS specifically requires the operator to verify that the required non-destructive testing (NDT) of the additional components has been completed before placing them into service during reconfiguration.

This ensures structural integrity and prevents the use of components with hidden defects.

WHY THE OTHER CHOICES ARE WRONG

- A** Magnetic inspection of every tower bolt is not required.
- C** Annual inspection does not apply to individual additional components before they are placed into service.
- D** Daily inspection by the prime contractor is not the RSOS requirement for reconfiguration components.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists 14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.04 Reconfiguration – Required knowledge includes inspections and NDT verification of additional components before they are placed into service.



BCCSA COR® Audit – Cranes
Verify that reconfiguration activities follow manufacturer's instructions and that NDT requirements are met.



KEY POINT: Before additional components are placed into service during reconfiguration, verify that the required NDT of those components has been completed.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 22 – ANSWER

QUESTION 22

A tower crane has been reconfigured. What must be done with the LMI and applicable limits before operation in the new configuration?

- A** Leave the previous settings until the next annual inspection
- B** Reduce all displayed capacities by 20%
- C** Reconfigure the LMI and verify and test the applicable limits for the new configuration 
- D** Reset only the maximum-radius limit

AFTER RECONFIGURATION – LMI AND LIMITS

After any reconfiguration (climbing, jib change, counter-jib change, component change, etc.), the LMI and all applicable limits must be set up for the new configuration and verified.



The LMI must be reconfigured and all applicable limits must be verified and tested before the crane is put into service.

WHAT THIS INCLUDES

RECONFIGURE THE LMI



Enter the correct configuration parameters (jib length, jib offset, counter-jib, tower height, etc.).

VERIFY APPLICABLE LIMITS



Check all limits that apply to the new configuration (e.g., radius, height under hook, slewing, load moment).

TEST LIMIT FUNCTIONS



Function test all limit devices to ensure they operate correctly and stop the crane as required.

CONFIRM CORRECT OPERATION



Ensure the LMI displays the correct information and the crane operates within the limits for the new configuration.

DOCUMENT AS REQUIRED



Record the verification and testing in the log as required by the manufacturer and company procedures.



Do not operate the crane in the new configuration until the LMI and all applicable limits have been verified and tested.



CORRECT ANSWER: C
RECONFIGURE THE LMI AND VERIFY AND TEST THE APPLICABLE LIMITS FOR THE NEW CONFIGURATION

After a reconfiguration, the LMI must be set up for the new configuration and all applicable limits must be verified and tested before the crane is put into service.

This ensures the crane operates safely within the limits for the new setup.

WHY THE OTHER CHOICES ARE WRONG

- A** Previous settings may be incorrect and create unsafe conditions.
- B** Arbitrarily reducing capacities is not permitted and may not provide adequate safety.
- D** Only resetting one limit does not ensure all applicable limits are set and functioning for the new configuration.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists 14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.04 Reconfiguration – Required knowledge includes verifying and testing LMI and limit devices for the new configuration before putting the crane into service.



BCCSA COR® Audit – Cranes
Verify that reconfiguration activities follow manufacturer's instructions and that LMI and limits are verified and tested before operation.



KEY POINT: After reconfiguration, the LMI must be reconfigured and all applicable limits verified and tested before the crane is operated in the new configuration.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 23 – ANSWER

QUESTION 23

Before tower crane reconfiguration begins, which combination must be verified?

- A** The configuration is acceptable to the manufacturer or professional engineer, and required permits have been obtained 
- B** Operator seniority and the project schedule
- C** The previous load chart and last daily inspection only
- D** Hook height and concrete strength only

VERIFICATIONS BEFORE RECONFIGURATION BEGINS

Before any reconfiguration work begins, the operator must verify that the proposed configuration is **acceptable to the manufacturer or a professional engineer**, and that all required permits/approvals have been obtained.

THE REQUIRED COMBINATION

1. ACCEPTABLE CONFIGURATION

The proposed reconfiguration (climbing, jib change, counter-jib change, component change, etc.) must be acceptable to:

- The crane manufacturer, OR
- A professional engineer (as required by the crane owner or project requirements)



2. REQUIRED PERMITS OBTAINED

All permits, approvals and notifications required by law, regulation, the crane owner and the site must be obtained before reconfiguration begins.



WorkSafeBC /
Regulatory
requirements



Municipal / Local
authorizations
(if applicable)



Crane owner /
Project permits
and approvals



Do not begin any reconfiguration work until the configuration is approved by the manufacturer or professional engineer AND all required permits are in place.



Beginning reconfiguration without both approvals and permits may result in serious injury, equipment damage, or regulatory violations.



CORRECT ANSWER: A

THE CONFIGURATION MUST BE ACCEPTABLE TO THE MANUFACTURER OR PROFESSIONAL ENGINEER, AND REQUIRED PERMITS MUST BE OBTAINED

Before reconfiguration begins, the operator must verify that the proposed configuration is acceptable to the manufacturer or professional engineer, and that all required permits and approvals have been obtained. This ensures the reconfiguration is safe, compliant and authorized.

WHY THE OTHER CHOICES ARE WRONG

- B** Operator seniority and project schedule are not safety or regulatory verification requirements.
- C** The previous load chart and last daily inspection alone do not confirm the new configuration is acceptable or authorized.
- D** Hook height and concrete strength are not the required verifications before reconfiguration.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.03 Reconfiguration – Required knowledge includes ensuring proposed configurations are acceptable to the manufacturer or a professional engineer and that required permits are obtained before reconfiguration begins.



BCCSA COR® Audit – Cranes
Verify that reconfiguration activities follow manufacturer's instructions and applicable legal and project requirements.



KEY POINT: Before reconfiguration begins, verify that the proposed configuration is acceptable to the manufacturer or professional engineer, and that all required permits and approvals are in place.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 23 – ANSWER

QUESTION 23

Before tower crane reconfiguration begins, which combination must be verified?

- A** The configuration is acceptable to the manufacturer or professional engineer, and required permits have been obtained 
- B** Operator seniority and the project schedule
- C** The previous load chart and last daily inspection only
- D** Hook height and concrete strength only

VERIFICATIONS BEFORE RECONFIGURATION BEGINS

Before any reconfiguration work begins, the operator must verify that the proposed configuration is **acceptable to the manufacturer or a professional engineer**, and that all required permits/approvals have been obtained.

THE REQUIRED COMBINATION

1. ACCEPTABLE CONFIGURATION

The proposed reconfiguration (climbing, jib change, counter-jib change, component change, etc.) must be acceptable to:

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Regulatory
requirements



Municipal / Local
authorizations
(if applicable)



Crane owner /
Project permits
and approvals



Do not begin any reconfiguration work until the configuration is approved by the manufacturer or professional engineer AND all required permits are in place.



Beginning reconfiguration without both approvals and permits may result in serious injury, equipment damage, or regulatory violations.



CORRECT ANSWER: A

THE CONFIGURATION MUST BE ACCEPTABLE TO THE MANUFACTURER OR PROFESSIONAL ENGINEER, AND REQUIRED PERMITS MUST BE OBTAINED

Before reconfiguration begins, the operator must verify that the proposed configuration is acceptable to the manufacturer or professional engineer, and that all required permits and approvals have been obtained. This ensures the reconfiguration is safe, compliant and authorized.

WHY THE OTHER CHOICES ARE WRONG

- B** Operator seniority and project schedule are not safety or regulatory verification requirements.
- C** The previous load chart and last daily inspection alone do not confirm the new configuration is acceptable or authorized.
- D** Hook height and concrete strength are not the required verifications before reconfiguration.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.03 Reconfiguration – Required knowledge includes ensuring proposed configurations are acceptable to the manufacturer or a professional engineer and that required permits are obtained before reconfiguration begins.



BCCSA COR® Audit – Cranes
Verify that reconfiguration activities follow manufacturer's instructions and applicable legal and project requirements.



KEY POINT: Before reconfiguration begins, verify that the proposed configuration is acceptable to the manufacturer or professional engineer, and that all required permits and approvals are in place.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 24 – ANSWER

QUESTION 24

After reconfiguration, the LMI still displays the crane's previous jib configuration. What should the operator do?

- A** Operate at 50% of displayed capacity
- B** Use the paper load chart and ignore the LMI
- C** Do not operate until the LMI is correctly reconfigured and applicable limits are verified and tested 
- D** Continue if no load is on the hook

LMI DISPLAYS PREVIOUS CONFIGURATION – DO NOT OPERATE

If the LMI still shows the previous jib configuration after reconfiguration, the displayed capacities and limits are incorrect for the current setup. Operating the crane under these conditions is unsafe and not permitted.



DO NOT OPERATE until the LMI is correctly reconfigured and all applicable limits are verified and tested.

LMI SHOWING PREVIOUS CONFIGURATION



INCORRECT
for the new configuration. Capacities and limits are not valid.

LMI CORRECTLY RECONFIGURED



Correct for the new configuration. Safe to operate.



All applicable limits must be reconfigured, verified, and tested before the crane is returned to service.



Operating with an incorrect LMI configuration can lead to overloaded crane conditions, structural failure, tip-over, serious injury, or death.



CORRECT ANSWER: C

DO NOT OPERATE UNTIL THE LMI IS CORRECTLY RECONFIGURED AND APPLICABLE LIMITS ARE VERIFIED AND TESTED

The operator must not operate the crane until the LMI is correctly set up for the new configuration and all applicable limits have been verified and tested.

This ensures the crane operates safely within the limits for the new setup.

WHY THE OTHER CHOICES ARE WRONG

- A** Operating at 50% of displayed capacity is not permitted and does not address the incorrect limits.
- B** The paper load chart does not replace the LMI and does not ensure limit functions are correct.
- D** Operating with no load is still operating. Hazards exist even without a load on the hook.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.03 Reconfiguration – Required knowledge includes verifying and testing LMI and limit devices for the new configuration before putting the crane into service.



BCCSA COR® Audit – Cranes
Verify that reconfiguration activities follow manufacturer's instructions and that LMI and limits are verified and tested before operation.



KEY POINT: Do not operate the crane until the LMI is correctly reconfigured for the new configuration and all applicable limits are verified and tested.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 25 – ANSWER

QUESTION 25

A top climb is scheduled. Wind remains below the manufacturer's maximum climbing limit, but gusts are increasing rapidly.

What is the **BEST** action?

- A** Begin immediately before the wind becomes stronger
- B** Evaluate the changing conditions and do not proceed unless the operation can remain within applicable limits and safe procedures 
- C** Increase climbing speed
- D** Use only the average wind reading

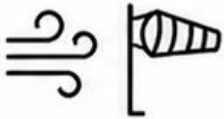
BEST ACTION: EVALUATE AND DO NOT PROCEED UNLESS SAFE

Even though current wind speed is below the manufacturer's maximum climbing limit, rapidly increasing gusts create a high risk.

The best action is to evaluate the changing conditions and do not proceed unless the operation can remain within applicable limits and safe procedures.

WHY THIS IS THE BEST ACTION

WIND CAN CHANGE QUICKLY



Gusts can exceed limits suddenly, creating hazardous conditions during the climb.

CLIMBING IS HIGH RISK



Top climbing reduces stability margins. Safety depends on staying within all limits.

ASSESS BEFORE YOU ACT



Check current wind speed, gusts, trend, and forecast. Ensure the climb can be completed within limits.

SAFETY OVER SCHEDULE



It is always better to wait or postpone than to risk a loss of control, collision, or structural failure.



EVALUATE BEFORE PROCEEDING

- Monitor wind speed, gusts, and trend
- Confirm the climb can be completed within the manufacturer's limits
- Follow the manufacturer's and site procedures
- If conditions are not stable and within limits – **STOP** and wait



CORRECT ANSWER: B

EVALUATE THE CHANGING CONDITIONS AND DO NOT PROCEED UNLESS THE OPERATION CAN REMAIN WITHIN APPLICABLE LIMITS AND SAFE PROCEDURES

This is the best action because it manages the risk from increasing gusts and ensures the climb can be completed safely within the limits.

Safety must never be compromised to stay on schedule.

WHY THE OTHER CHOICES ARE WRONG

- A** Starting immediately ignores the rapidly increasing gusts and the risk of exceeding limits during the climb.
- C** Increasing climbing speed increases load and dynamic forces and does not reduce wind risk.
- D** Using only the average wind reading ignores gusts, which are critical for crane safety.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists 14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.03 Top Climbing – Required knowledge includes wind effects and operating within manufacturer's climbing limits.



BCCSA COR® Audit – Cranes
Verify that top climbing is planned and performed in accordance with manufacturer's instructions and within wind limits.



KEY POINT: Do not proceed with a top climb when gusts are increasing. Evaluate the conditions and only proceed when the operation can remain within applicable limits and all safe procedures.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 26 – ANSWER

QUESTION 26

During a top climb, a mast section does not align correctly with its required connection points.

What should the crew do?

- A** Use the hydraulic climbing system to force the section into alignment
- B** Install the connections that align and correct the others later
- C** Stop, determine the cause and follow the applicable manufacturer's procedure 
- D** Strike the section until it aligns

BEST ACTION: STOP, INVESTIGATE AND FOLLOW PROCEDURE

If a mast section does not align correctly, forcing or partially installing it can damage components, create unsafe stresses, or cause structural failure.

The crew must stop, determine the cause, and follow the applicable manufacturer's procedure before continuing the climb.

WHY STOP AND INVESTIGATE?

PREVENTS DAMAGE AND FAILURE



Forcing or striking can damage alignment pins, holes, or the climbing frame and lead to structural failure.

IDENTIFIES THE CAUSE



Misalignment can be caused by debris, paint build-up, distortion, or incorrect positioning.

FOLLOWS MANUFACTURER'S PROCEDURE



The manufacturer's procedure provides the correct steps to resolve the issue safely.

MAINTAINS SAFE CLIMBING PRACTICES



Stopping and correcting the issue ensures the crane remains stable and the climb is completed safely.



NEVER FORCE, PARTIALLY INSTALL, OR STRIKE COMPONENTS.
These actions can compromise the crane, the climbing system, and the safety of personnel.



CORRECT ANSWER: C

STOP, DETERMINE THE CAUSE AND FOLLOW THE APPLICABLE MANUFACTURER'S PROCEDURE

Stopping and investigating the cause of misalignment and following the manufacturer's procedure ensures the climbing system and mast sections are not damaged and the operation is completed safely and in accordance with the manufacturer's instructions.

WHY THE OTHER CHOICES ARE WRONG

- A** Forcing the section can damage components, overload the system, and create a serious safety hazard.
- B** Partially installing connections can lead to misalignment, overstress, and failure during or after the climb.
- D** Striking the section can damage pins, holes, and structures, and can cause the section to shift unexpectedly, creating a severe hazard.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.03 Top Climbing – Required knowledge includes following the manufacturer's instructions and procedures for climbing operations.



BCCSA COR® Audit – Cranes
Verify that top climbing activities follow manufacturer's instructions and that misalignment or faults are addressed in accordance with those instructions before continuing.



KEY POINT: If a mast section does not align, STOP THE CLIMB. Determine the cause and follow the manufacturer's procedure before proceeding.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 27 – ANSWER

QUESTION 27

A climbing stage has been completed, but one required structural connection cannot be verified. The crew is ready for the next stage. What should happen?

- A** Continue and verify the connection afterward
- B** Do not continue until the required connection is correctly completed and verified
- C** Continue while the hydraulic system carries the load
- D** Have the operator visually check the connection from the cab

BEST ACTION: DO NOT CONTINUE UNTIL VERIFIED

All required structural connections must be correctly completed and verified before the next climbing stage begins. Proceeding without verification can lead to loss of stability, structural failure, or collapse.



Do not continue until the required connection is correctly completed and verified.

WHY THIS IS THE RIGHT ACTION

ENSURES STRUCTURAL INTEGRITY



All connections are essential to maintain the strength and stability of the crane tower.

PREVENTS CATASTROPHIC FAILURE



An unverified or improper connection can lead to collapse, equipment damage, serious injury, or death.

FOLLOWS MANUFACTURER'S PROCEDURE



The manufacturer's instructions require all connections to be completed and verified before continuing.

SAFE WORK PRACTICE



Safe climbing depends on discipline, patience, and following the procedure every time.



NEVER PROCEED TO THE NEXT STAGE UNTIL EVERY REQUIRED CONNECTION IS CORRECTLY COMPLETED AND VERIFIED.
It is not safe, it is not permitted, and it is not worth the risk.



CORRECT ANSWER: B

DO NOT CONTINUE UNTIL THE REQUIRED CONNECTION IS CORRECTLY COMPLETED AND VERIFIED

Proceeding without verification puts the stability of the tower at risk. All connections must be confirmed in accordance with the manufacturer's procedure before the next stage begins.

Verification ensures the crane remains safe for the crew and the load.

WHY THE OTHER CHOICES ARE WRONG

- A** Verifying later does not eliminate the risk during the next stage.
- C** The hydraulic climbing system is not a substitute for structural connections. It is not designed to carry load if connections are incomplete or unverified.
- D** Visual checks from the cab are not a valid verification method. Connections must be physically inspected and verified by qualified personnel at the connection points.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.03 Top Climbing – Required knowledge includes ensuring structural connections are installed and verified in accordance with the manufacturer's instructions before continuing.



BCCSA COR® Audit – Cranes
Verify that top climbing activities follow manufacturer's instructions and that all connections are completed and verified before proceeding.



KEY POINT: Do not continue the climb until every required structural connection is correctly completed and verified in accordance with the manufacturer's procedure.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 28 – ANSWER

QUESTION 28

A qualified climbing supervisor proposes changing the manufacturer's climbing sequence to save time. The change is not covered by the manufacturer's instructions.

What is the BEST response?

- A** Follow the supervisor because experience overrides the manual
- B** Proceed if the entire climbing crew agrees
- C** Do not deviate from the applicable instructions unless the variation is properly addressed by professional-engineer instructions as required 
- D** Test the modified sequence using reduced hydraulic pressure

BEST RESPONSE: FOLLOW THE APPLICABLE INSTRUCTIONS

Manufacturers design climbing sequences based on engineering, testing, and safety factors. Deviating from these instructions without proper engineering approval can create unforeseen risks, compromise structural integrity, and void certifications.



Do not deviate from the applicable instructions unless the variation is properly addressed by professional-engineer instructions as required.

WHY THIS IS THE BEST RESPONSE

RESPECTS ENGINEERING AND DESIGN



Manufacturer instructions are based on engineering, testing, and certification requirements.

MAINTAINS SAFETY AND STABILITY



Following the approved sequence maintains structural integrity and the stability of the crane during the climb.

MEETS REGULATORY REQUIREMENTS



Deviations without engineering approval do not comply with regulations and can create legal liability.

AVOIDS SERIOUS CONSEQUENCES



Improper sequences can cause component failure, overload, collapse, injury, or death.



ONLY DEVIATE FROM THE MANUFACTURER'S INSTRUCTIONS WHEN A PROFESSIONAL ENGINEER HAS PROVIDED WRITTEN INSTRUCTIONS ADDRESSING THE VARIATION AS REQUIRED.



CORRECT ANSWER: C

DO NOT DEVIATE FROM THE APPLICABLE INSTRUCTIONS UNLESS THE VARIATION IS PROPERLY ADDRESSED BY PROFESSIONAL-ENGINEER INSTRUCTIONS AS REQUIRED

The manufacturer's climbing sequence is based on engineering, testing, and safety factors. Deviating from these instructions without professional-engineer approval introduces unquantified risks and can compromise the crane, the climb, and the safety of personnel and the load.

Only a professional engineer can evaluate the proposed change and issue written instructions addressing the variation, if it is acceptable.

WHY THE OTHER CHOICES ARE WRONG

- A** Experience is valuable, but it does not override engineered instructions and safety requirements.
- B** Crew agreement does not make an unapproved deviation safe or compliant.
- D** Testing a modified sequence, even at reduced pressure, is unsafe and not permitted. It can still cause damage or collapse.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists
14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.03 Top Climbing – Climbing must be carried out in accordance with the manufacturer's instructions.



BCCSA COR® Audit – Cranes
Verify that top climbing activities follow manufacturer's instructions and that deviations are only made where professional-engineer instructions address the variation.



KEY POINT: Do not deviate from the manufacturer's climbing sequence unless the variation is properly addressed by professional-engineer instructions as required.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 29 – ANSWER

QUESTION 29

During a climbing operation, required radio communication between the operator and climbing crew is lost. What should happen?

- A** Continue using hand signals even though the entire crew cannot maintain visual contact
- B** Complete the current climbing stroke before stopping
- C** Stop the operation safely until the required communication is restored 
- D** Continue if the qualified supervisor remains on the crane

BEST ACTION: STOP UNTIL COMMUNICATION IS RESTORED

Effective communication is critical during climbing. Loss of required radio communication removes a key layer of control and awareness. The safest action is to stop the operation immediately and restore communication before continuing.



Stop the operation safely until the required communication is restored.

WHY THIS IS THE RIGHT ACTION

PREVENTS MISCOMMUNICATION



Loss of radio contact greatly increases the risk of misunderstanding and unsafe actions.

MAINTAINS CONTROL OF THE OPERATION



Stopping ensures the operator and crew maintain full control and awareness.

PROTECTS PEOPLE, EQUIPMENT & STRUCTURE



Stopping prevents incidents such as dropped loads, structural damage, or tipping.

FOLLOWS SAFE WORK PRACTICES



Safe procedures require communication to be confirmed and maintained at all times.



COMMUNICATION IS A CRITICAL CONTROL.

If it is lost – STOP the climb, restore communication, and confirm all personnel are ready before continuing.



CORRECT ANSWER: C

STOP THE OPERATION SAFELY UNTIL THE REQUIRED COMMUNICATION IS RESTORED

Climbing operations rely on clear, continuous communication. If required radio communication is lost, the operator cannot safely coordinate movements or confirm crew actions.

Stop the operation in a safe state and restore communication before proceeding.

WHY THE OTHER CHOICES ARE WRONG

- A** Hand signals require full visual contact. Without it, miscommunication can occur and lead to serious incidents.
- B** Completing a stroke without communication increases risk and removes the ability to stop if a problem is identified.
- D** A supervisor's presence does not replace required communication. The operator must maintain direct communication with the crew during all climbing steps.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists 14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023) C-7.03 Top Climbing – Effective communication must be maintained between the operator and climbing crew during all climbing activities.



BCCSA COR® Audit – Cranes Verify that top climbing activities follow manufacturer's instructions and that communication is maintained throughout the climb.



KEY POINT: If required communication is lost during a climb, STOP the operation safely. Restore communication and confirm the crew is ready before continuing.



TASK C-7: CLIMBING AND RECONFIGURATION

BC RED SEAL TOWER CRANE OPERATOR – PRACTICE TEST

QUESTION 30 – ANSWER

QUESTION 30

The physical climbing operation is complete and the crane appears normal. Lifting operations are waiting. What is the **BEST** action?

- A** Resume lifting below 50% capacity
- B** Perform one test lift and return to normal operation
- C** Complete the required post-climb verification, applicable certifications, adjustments and documentation before returning the crane to service 
- D** Allow the operator to determine whether additional verification is necessary

BEST ACTION: COMPLETE REQUIRED POST-CLIMB VERIFICATION

Even if the crane appears normal, the climbing process affects critical structural components, connections, limits and settings.

The best action is to complete all required verification, certifications, adjustments and documentation before returning the crane to service.

WHY THIS IS THE BEST ACTION

CONFIRMS STRUCTURAL INTEGRITY



Verification ensures all connections, pins, bolts and structural components are correct and secure.

VERIFIES SYSTEMS AND LIMITS



All limits, functions and safety systems must be checked, set and tested for the new configuration.

MEETS REGULATORY REQUIREMENTS



Applicable certifications and records are required by the manufacturer, CSA standards and OHS regulations.

ENSURES SAFE OPERATION



Completing these steps confirms the crane is safe for lifting and protects people, equipment and the structure.

PROVIDES DOCUMENTATION



Proper documentation creates a record of compliance and supports future inspections and maintenance.



DO NOT RETURN THE CRANE TO SERVICE UNTIL ALL POST-CLIMB REQUIREMENTS ARE COMPLETED AND DOCUMENTED.
Never assume the crane is ready based on appearance alone.



CORRECT ANSWER: C

COMPLETE THE REQUIRED POST-CLIMB VERIFICATION, APPLICABLE CERTIFICATIONS, ADJUSTMENTS AND DOCUMENTATION BEFORE RETURNING THE CRANE TO SERVICE

Post-climb verification confirms the crane is safe and compliant for the new configuration.

This includes structural checks, limit testing, system verification, adjustments and documented certification as required by the manufacturer and regulations.

Lifting operations must wait until the crane is officially returned to service.

WHY THE OTHER CHOICES ARE WRONG

- A** Operating at reduced capacity without verification does not eliminate the risk of structural or system failure.
- B** A single test lift does not confirm all systems, limits and structural connections are correct and safe.
- D** Decisions about required verification are not left to the operator. They must follow the manufacturer's instructions and regulatory requirements.

REFERENCE BASIS



WorkSafeBC OHS Regulation, Part 14 – Cranes and Hoists 14.80(1)(a) A crane must be operated in a manner that ensures stability.



Red Seal RSOS (2023)
C-7.03 Top Climbing – Post-climb verification, testing, certification and documentation must be completed before the crane may be returned to service.



BCCSA COR® Audit – Cranes
Verify that top climbing activities are followed by complete verification, testing, certifications, adjustments and documentation before the crane returns to service.



KEY POINT: The crane is not ready for lifting until post-climb verification, certifications, adjustments and documentation are complete and the crane is returned to service.